

Thomson Reuters GFMS

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Quarterly Three Year PGMs Forecast

Prepared by Thomson Reuters GFMS

October 2011



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This report forms Thomson Reuters GFMS latest price forecast for platinum and palladium and covers the three year period 2012-2014. It will be updated on a quarterly basis, the next edition being scheduled for release early in the first quarter. Similar forecasts are available for gold and silver. To obtain further detail on these products, please contact Charles de Meester, tel: +44 (0) 7500 834 954, email: charles.demeester@thomsonreuters.com.

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Table of Contents

1.0 Executive Summary	4
2.0 Economic Assumptions	9
3.0 Mine Production	13
4.0 Autocatalyst Scrap	17
5.0 Above-ground Stocks	19
6.0 Fabrication	22
- <i>autocatalyst</i>	22
- <i>jewellery</i>	26
- <i>industrial</i>	31
7.0 Investment	34



Notes

Units used:

Unless otherwise stated, all statistics on platinum and palladium supply and demand are expressed in terms of fine metal content. All references in this publication to "ounces" refer to troy ounces.

One troy ounce =	31.103 grammes
One tonne =	32,151 troy ounces
"dollar"	US dollar unless otherwise stated

Prices:

Unless otherwise stated, US dollars and their equivalents are for the pm fixes of the London Platinum and Palladium Market.

Terminology:

Estimates of **supply** include mine production and the recycling of both scrapped autocatalysts and old jewellery but exclude contributions from above-ground stocks, such as supplies from stocks controlled by state institutions in Russia.

Demand estimates are *net* of recycling with the exception of autocatalyst and jewellery, where *gross* demand is shown. i.e. the total amount of metal supplied to these two sectors. Estimates of recycling from scrapped autocatalysts and jewellery are shown separately as part of supply given their scale and potential for change. (This year's survey is the second in which we have presented jewellery fabrication as gross with a separate series for scrap; prior to this, jewellery was shown on a net basis.) Estimates of demand exclude the movements of any above-ground stocks held within the specified industries, for example any changes in stocks held by the automotive industry.

By simple arithmetic, this leaves either a **"Gross Surplus or Deficit"** before any movements in above-ground stocks are considered. This is a critical measure of the underlying fundamentals of platinum and palladium and indicates the extent to which fabrication demand may have depended on the release of above-ground stocks, or otherwise. At the same time, this also indicates the change in global above-ground stocks.

Unless otherwise stated, all references to **"above-ground stocks"** of platinum and palladium refer to stocks of refined metal, of a form and quality accepted as good delivery in the Zurich market and the world's principal commodity exchanges. Our supply/demand tables also show **"Estimated Movements in Stocks"**. These specific movements relate only to above-ground stock holdings for which reasonable estimates of movement can be made and attributed. A listing and breakdown of these appears in the more detailed tables in the Appendices section of this Survey.

Having allowed for the Estimated Movements in Stocks as defined above, the **"Residual Surplus or Deficit"** is arrived at by deduction. A **residual deficit** implies the extent to which **other above-ground stocks**, including those held by financial institutions and/or investors, were released to meet fabrication demand. Conversely a **residual surplus** implies the extent to which these **other above-ground stock** holdings were added to. However, this should not be construed as indicating the change in global above-ground stocks. For this, please refer to the reported Gross Surplus or Deficit.

Throughout the tables, totals may not add due to independent rounding.



1. Executive Summary

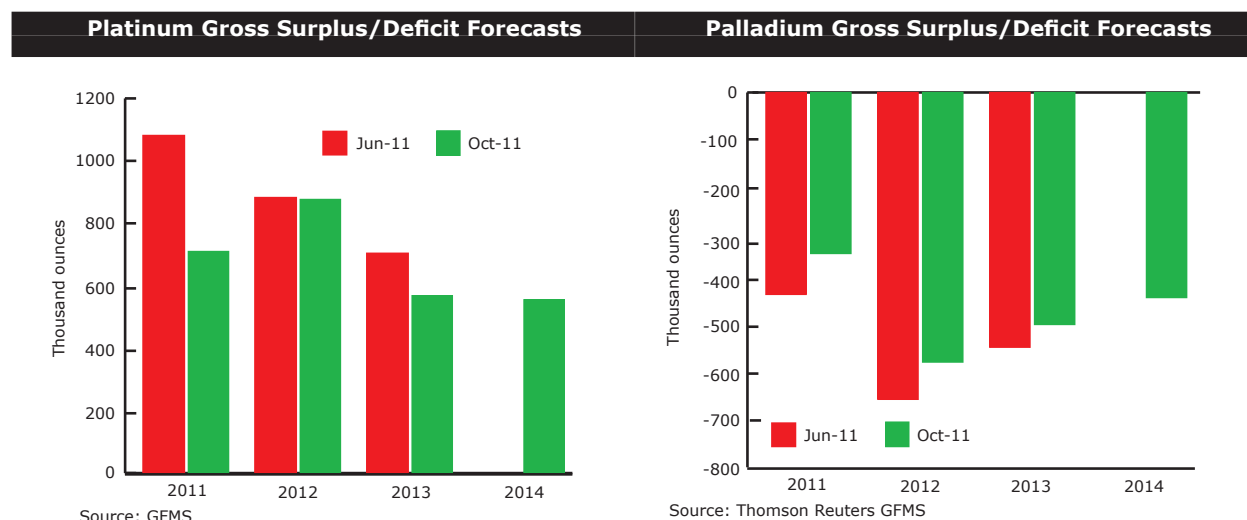
As far as platinum and palladium prices are concerned, much has changed since our previous report in June of this year – specifically, over recent weeks since late September. However, while the current price environment is materially different, our view of the underlying fundamentals is largely unchanged. After another significant gross surplus this year (0.7m ounces), platinum is expected to remain in surplus for the duration of the forecast period, although the scale of these is expected to ease somewhat in 2013-14. Meanwhile, 2011 should result in a further although reduced gross deficit in palladium (0.34m ounces), with continuing shortfalls anticipated in each of the forecast years thereafter; these being somewhat larger than the level we expect this year. (Note: these underlying balances exclude stock related activity such as ETF flows and Russian government inventory sales).

Inevitably, these underlying balances have a direct impact on above-ground stocks. In the case of platinum, this results in a continuing escalation, while global stocks of palladium (the more abundant of the two) are expected to erode. Using our stock estimates for the end of last year as a reference (3.8m and 11.4m ounces respectively), these predicted balances will elevate above-ground platinum stocks to an estimated 6.5m ounces by the end of 2014, with stocks of palladium declining to a total of 9.6m ounces. Over the longer term, the negative implications of this for platinum are obvious, as are the more positive connotations for palladium.

On the whole, these outcomes for platinum and palladium are symptomatic of their relative fortunes

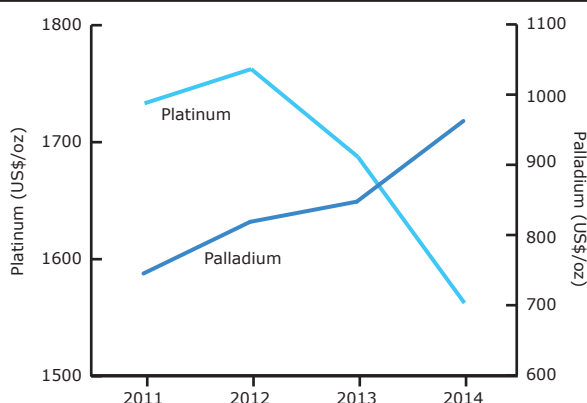
in the all important autocatalyst sector, this being the largest source of demand for both metals. Dominant in diesel applications, platinum is heavily exposed to Europe; this region being the world’s largest light diesel market and hence the biggest consumer of platinum in autocatalysts. As well as lacklustre vehicle production in Europe, platinum faces additional headwinds due to the absence of market share growth for light diesels and continuing substitution by palladium. Therefore, with a lack of momentum from its principal source of demand, platinum’s forecast growth in autocatalyst is entirely due to rising demand for heavy duty diesel applications, including the emerging off-road segment. However, exciting as this undoubtedly is, it will not be sufficient to restore platinum’s vibrancy in the autocatalyst sector as a whole, with demand in 2014 still distant from the high of 2007.

Meanwhile, palladium is even more dominant in light gasoline applications (where its strangle-hold is increasing), which comprise the vast majority of global vehicle production. This gives palladium a more diverse geographic footprint, which includes greater exposure to growth markets such as China and the emerging economies where emissions standards are also tightening. Furthermore, demand for palladium is also assisted by continuing substitution gains versus platinum in both diesel and gasoline applications, even though these are expected to slow in comparison with recent years. Consequently, despite the economic woes in Europe and the United States, we expect record high demand for palladium in autocatalysts as early as next year, with further gains anticipated over the rest of the forecast period.





Platinum & Palladium Price Forecast



Source: Thomson Reuters GFMS

However, in the medium term (the next 18 months or so), the overarching climate for precious metals investment is expected to remain the dominant force as far as prices are concerned. When our last report was released in June, platinum and palladium traded at averages that month of \$1,770 and \$771 respectively and averaged not far from these levels through to mid-September. From that point, however, amid a widespread sell-off in commodities and equities, platinum touched a near two year low of \$1,442 in early October trading, with palladium sinking to \$549 – its lowest level in more than a year. Although both have since managed something of an improvement, this has been hesitant and lacking in conviction thus far, with prices still well adrift of recent levels.

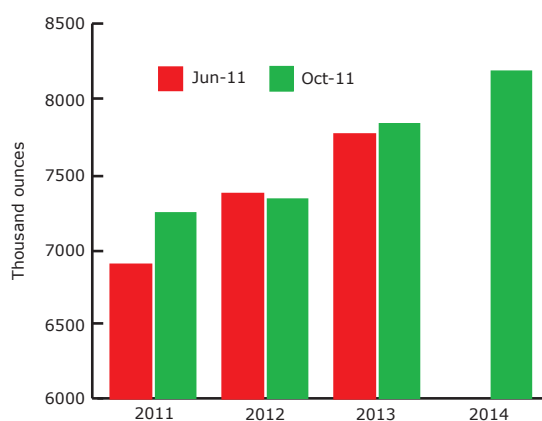
In the main, the causes of this sell-off were centred on the deepening sovereign debt crisis in the eurozone and the associated squeeze on liquidity. Amid intense anxiety pending a credible plan to manage a default by Greece, recapitalise the region's banks and prevent further contagion (and indeed, save the euro!), the economic recovery in Europe has stalled and the danger

of sliding back into recession is mounting. Meanwhile, the recovery in the US has also lost momentum in the face of continuing high unemployment and a depressed housing market, with already weak consumer confidence having been further dented by the recent debt-ceiling fiasco.

By late September, in the midst of this turmoil, action from the Fed to help revive a flagging US economy turned out to be a reprise of "Operation Twist" from the 1960s (a measure to curb long-term interest rates). This policy choice was preferred over a third round of quantitative easing which, given its inflationary implications, would have been very favourable for precious metals (gold in particular) had it occurred. As it turned out, with liquidity drying up, amid rising anxiety over eurozone exposures, this preference against printing more money, sparked fears of an impending cash-crunch similar to that seen in late 2008, triggering the sell-off of riskier assets and a flight to cash.

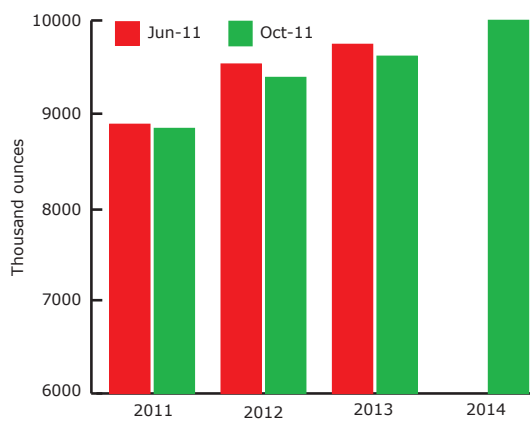
While brutal for platinum and palladium (with gold taking big hits as well), this has thus far proved much less extreme than the bail-out of late 2008. On the demand side of the equation, platinum's slide has been arrested by positive responses from jewellery fabrication in China and retail investment in Japan, helped in both cases by platinum being less expensive than gold (an exceptionally rare event). Meanwhile, unlike the situation in 2008, the cost of mine production in South Africa (of which there is greater awareness) appears to have come into play much earlier than before, encouraging bargain-hunting investors back into platinum and helping to support prices.

Platinum Demand Forecast Revisions



Source: Thomson Reuters GFMS

Palladium Demand Forecast Revisions



Source: Thomson Reuters GFMS



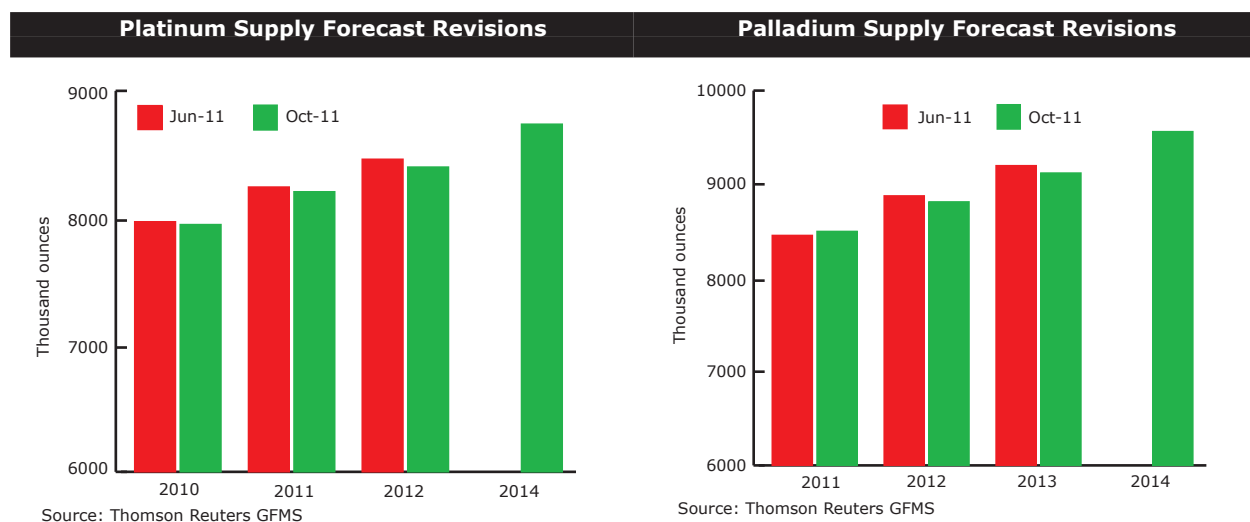
Simultaneously, palladium has also found a similar degree of support, most likely aided by the interruption of platinum's decline, with the long established differential between the two metals having been preserved in the vicinity of \$1,000 throughout this period (although palladium's losses in percentage terms have been relatively more severe). While this support may be attributable in part to South African mine economics as well, palladium's re-engagement with investors is most likely due to its promising underlying fundamentals and the anticipated conclusion before long of Russian government inventory sales.

With their recent losses now stemmed, we believe that platinum and palladium prices will enter a period of rehabilitation over the remainder of this year, regaining some but not all of their recent losses. For 2011 as a whole, we forecast average prices of \$1,733 and \$745 respectively; these in fact being record highs, despite a much weaker fourth quarter. This partial recovery in the short term owes much to the continuing uncertainties over eurozone policy and the prospects for economic growth in Europe and the US next year. In contrast, we are considerably more positive about gold, which we expect to quickly recoup its recent losses before going on to thrive next year in a volatile atmosphere of latent uncertainty. Indeed in our base case assumptions, we see gold peaking in 2012, with prices trading frequently above \$2,000 over the latter part of next year.

While such a climate may not be entirely conducive for platinum and palladium (due to their demand base in autocatalyst and industrial applications), we believe they will find renewed investor interest in this environment, with supply-side economics continuing to feature prominently. Consequently, we expect

somewhat stronger average prices in 2012 than those we anticipate this year, and forecast averages of \$1,763 and \$819 respectively. However, these gains are slender by comparison with the much stronger performance we expect from gold next year and are tempered by continuing uncertainties in the broader economy, especially in Europe and the US. Therefore, as a consequence of gold's anticipated strength next year, we expect platinum prices on average to remain below gold throughout 2012 and most likely into early 2013. This should be supportive of platinum jewellery demand next year, even though average platinum prices are expected to return to elevated levels.

Looking forward to 2013, this is the year in which we expect the bull market in gold to begin to dissipate, especially in the second half of the year. We see this as symptomatic of the eventual resolution of many of the high profile financial problems the world currently faces, the emergence of sustainable economic growth and a move towards positive real interest rates in western economies. While such conditions may initially appear favourable for both platinum and palladium, this will inevitably usher in a period of heavy investor selling, led by the significant liquidation of positions in gold. Given platinum's traditionally strong relationship with gold, we expect it to suffer more in this period than palladium, weighed down by its continuing gross surplus and swollen above-ground stocks. Meanwhile, although palladium may suffer some collateral damage, we believe investors will be more inclined to maintain their exposure, given what should by then be increasingly attractive fundamentals and with Russian government sales all but over. We forecast average prices in 2013 of \$1,688 and \$848 for platinum and palladium respectively, these denoting a material contraction in the price differential between them.





Finally, as we move into 2014 (and with gold continuing its decline), we believe platinum will ease somewhat further but remain supported at levels close to (but not always exceeding by much) the cost of mine production in South Africa. However, as with the price levels we forecast for 2013, this is likely to be very good for platinum jewellery demand, thus enabling fabrication demand in total (all applications) to reach a new all-time high in 2014, having equalled the 2007 peak the previous year. In the case of palladium, however, we see 2014 as the year when prices are at last a much fairer reflection of its improved underlying fundamentals. As it will have been throughout the forecast period, the driver will again be increasing autocatalyst demand which, together with other applications, should see fabrication demand for palladium at 10m ounces for the first time ever. We believe this should be sufficient to enable palladium to outperform both platinum and gold, with prices eventually consolidating at the \$1,000 level. Our forecast averages for 2014 are \$1,563 and \$963 for

platinum and palladium respectively, with the price differential narrowing still further.

Having referred several times to the cost of mine production in South Africa as supportive of platinum and to a lesser extent palladium prices, it is worth sounding a note of caution in this regard. While escalating costs suggest a necessity for commensurately higher prices (with production cuts threatened or materialising if sufficient incentive prices fail to emerge), this concept relies on either a reasonably close balance between new supply and fabrication demand, or that prices will soon force such a balance to quickly emerge. However, in the case of platinum, with a recent history and outlook of persistent underlying gross surpluses (and rising above-ground stocks), this is clearly not the case. Similarly, although palladium is in gross deficit and likely to remain so, there is clearly no shortage with above-ground palladium stocks remaining substantial, although declining, over the forecast period.

Platinum Supply/Demand					© Thomson Reuters GFMS				
(000 ounces)									
	2011	2012	2013	2014					
SUPPLY									
Mine production									
South Africa	4,717	4,908	4,980	5,144					
Russia	811	790	782	796					
North America	413	394	397	415					
Other	456	454	517	571					
Total mine production	6,397	6,546	6,676	6,927					
Autocatalyst scrap	986	1,106	1,197	1,290					
Old jewellery scrap	564	554	528	521					
Sub total - Supply	7,948	8,206	8,401	8,738					
DEMAND									
Autocatalysts	3,035	3,107	3,321	3,591					
Jewellery	2,072	2,193	2,425	2,704					
Chemical	463	464	556	491					
Electronics	238	262	250	247					
Glass	487	433	491	386					
Petroleum	200	253	186	156					
Retail investment	255	97	66	41					
Other industrial	488	521	537	567					
Sub total - Demand	7,238	7,330	7,832	8,183					
Gross surplus/(deficit)	710	876	568	555					
Identifiable stock movements									
Industry stocks	0	0	0	0					
Exchange Traded Funds*	(300)	(150)	200	100					
Sub total - stock movements	(300)	(150)	200	100					
Residual surplus/(deficit)	410	726	768	655					
Average price	\$1,733	\$1,763	\$1,688	\$1,563					
*negative values for ETFs are investment inflows, positive are outflows									



Palladium Supply/Demand				
© Thomson Reuters GFMS				
(000 ounces)				
	2011	2012	2013	2014
SUPPLY				
Mine production				
South Africa	2,651	2,787	2,818	2,892
Russia	2,715	2,668	2,647	2,661
North America	955	957	1,014	1,159
Other	544	545	589	630
Mine supply	6,864	6,957	7,067	7,343
Autocatalyst scrap	1,460	1,670	1,868	2,030
Old jewellery scrap	174	186	186	191
Sub total - Supply	8,498	8,813	9,121	9,563
DEMAND				
Autocatalysts	5,488	6,068	6,361	6,803
Jewellery	698	629	577	549
Dental	647	650	653	628
Chemical	361	373	372	371
Electronics	1,456	1,495	1,482	1,480
Retail investment	99	81	69	62
Other industrial	93	93	103	107
Sub total - Demand	8,842	9,389	9,616	10,001
Gross surplus/(deficit)	(344)	(575)	(495)	(438)
Identifiable stock movements				
Russia	800	500	200	0
Exchange Traded Funds*	300	(200)	(100)	300
Sub total - stock movements	1,100	300	100	300
Residual surplus/(deficit)	756	(275)	(395)	(138)
Average price	\$745	\$819	\$848	\$963
*negative values for ETFs are investment inflows, positive are outflows				



2. Economic Assumptions

2.1 Introduction

Thomson Reuters GFMS' views for the forecast period are presented in the form of three probability-weighted scenarios, namely the Base Case Scenario, Scenario B and Scenario C. Under our **Base Case**, we expect a slowdown in the world GDP growth this and next year. This is mainly driven by persistent sovereign debt crises, fiscal policy tightening and a slowdown in industrial production in major advanced economies. Although developing world countries continue to expand, global growth is likely to moderate due to tight monetary policies in a number of countries and downward pressure on external demand stemming from economic and financial problems in industrialised countries. However, we expect global growth to improve gradually over the final two years of our forecast which is mainly driven by emerging markets. We view this as the most likely scenario with a 45% probability.

Our more bullish **Scenario B** with a 40% probability predicts a global economic recovery with higher growth rates in the major advanced and developing countries over the forecast period. Under this scenario, we assume that the US government and the Federal Reserve continue to implement loose fiscal and monetary policies. As a consequence, we expect negative real interest rates and rising inflation, with the latter reaching an annual average of 5.8% in 2014. Moreover, the US dollar weakens under this scenario. It is of note that we have raised the possibility of Scenario B by 10% since our June forecast, reflecting increasing concerns over the sovereign debt problem on both sides of the Atlantic over the last couple of months.

Under our bearish **Scenario C**, to which we assign a 15% probability (down by 5% from the previous report), we expect the effect of government measures to be negligible on the longer run growth rate as the global economy struggles to escape deflation. Consequently, global GDP growth slows appreciably in 2011 and remains sluggish over much of the rest of the forecast period, particularly in the developed world.

The sections below start by describing the background to the current economic situation. We then discuss in more detail each of the three economic scenarios with a particular focus on some individual countries.

2.2 Economic Backdrop

Growing concerns over the sovereign debt problem in both Europe and the United States not only undermined the stability of banking systems and financial markets, but also raised worries regarding a potential slowdown in the world economy. Indeed, the IMF cut its predictions for GDP growth in Europe for the next two years. The outlook for Greece remains the worst as its economy is expected to contract for four successive years. Furthermore, Italy's credit rating has been recently downgraded by Standard & Poor's, Moody's, and Fitch on the back of its weakening growth prospects and structural economic problems. Similarly, Spanish government debt has been recently downgraded by all major credit rating agencies due to growing concerns about the country's ability to cut its debt and about its growth prospects.

As the European sovereign debt crisis intensifies, equity investors are becoming more reluctant to provide liquidity, putting pressure on European banks and the real economy. As a result, the shares of European banks have fallen by 40% over the last three months. The price of Credit Default Swaps (CDS), a type of insurance against default, also rose, as the banks are struggling to obtain funding.

The situation in Europe remains very complex and it is still very difficult to predict the outcome of the Eurozone debt problem. One of the possible solutions could be a rapid shift to a full fiscal union. This would put the Eurozone gross government debt to still high but manageable level. Nevertheless, we regard this as a very unlikely outcome as the member states would be highly reluctant to give up their fiscal independence. Another possible solution for Greece is to accept the present "adjustment programme" supervised by the

Real GDP Growth Rates - Base Case Scenario

© Thomson Reuters GFMS

(%)	2011	2012	2013	2014
World	2.8	2.5	3.3	3.5
United States	1.6	1.5	2.3	2.1
EU-27	1.5	1.0	1.9	2.1
Japan	-0.5	2.0	2.0	2.0
China	9.0	8.5	8.5	8.0
India	7.7	7.2	7.9	8.1

Source: Thomson Reuters GFMS, World Bank



European Commission, the European Central Bank and the International Monetary Fund, but at the expense of little prospect of growth. Finally, Greece's default and exit from the euro, accompanied by local currency depreciation, is another way out of the crisis which would restore the country's growth at a cost of losing its membership in the eurozone. It is also of note, that none of our scenarios include the Greek default.

Moving across the Atlantic, the United States avoided a near-term default as politicians eventually agreed on raising the US debt ceiling, which was earlier set at \$14.3 trillion. However, given concerns over budget deficits, Standard & Poor's cut the long-term US credit rating from 'AAA' to 'AA+' for the first time ever, leading to the largest fall in global stock markets in almost three years. The Federal Reserve has warned about the outlook for the US economy as the world's largest economy is struggling with huge federal debt, an elevated unemployment rate and rising fears of a possible double-dip recession. Likewise, the IMF has recently cut its forecast for the US growth to 1.5% this year, down from its June projection of 2.5%.

In addition, a number of developing economies continue to follow policy tightening in order to reduce inflationary pressures and strengthen fiscal accounts, also slowing the world economy. Moreover, concerns over "currency wars" and trade protectionism are rising as some countries are putting downward pressure on their currencies. Overall, the current economic backdrop remains favourable for investment in gold and, to an extent, all precious metals.

2.3 Base Case Scenario

Real **global** GDP growth is expected to ease back to 2.8% this year, having rebounded to 5.1% in 2010. This is due to the indebted nature of both governments and consumers in much of the developed world. Meanwhile, further tightening of monetary policies in emerging markets to fight surging inflation will also slow down world GDP growth this year. Next year, we expect a further decline in global GDP growth, an annual average of 2.5% due to a deterioration in the European sovereign debt problem. This is followed by faster growth in 2013 and 2014 as the more traditional sources of growth, such as consumer spending and the property market, start to improve. Both will be aided by unemployment beginning to slowly recede, albeit from elevated levels.

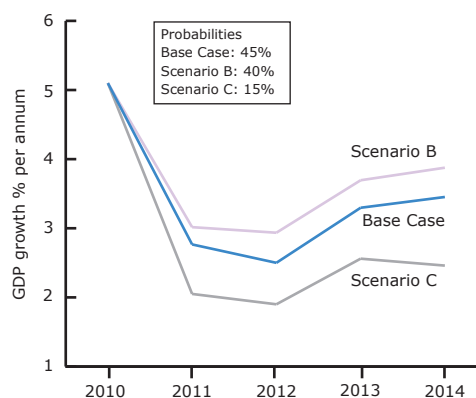
While **US** GDP growth for the second quarter of this year has been recently revised up to 1.3% from the

preliminary estimate of 1.0%, the economic recovery remains fragile. Central to this is ongoing weakness in consumer spending, which accounts for some 70% of American economic activity. Meanwhile, it also reflects a continuation of an elevated unemployment rate coupled with a lacklustre property market, both of which are likely to remain in place for an extended period. Indeed, facing a major slowdown in economic growth, the Federal Reserve has already confirmed that it would like to keep interest rates unchanged until mid-2013. Furthermore, a new stimulus plan known as "Operation Twist" has been recently announced. Under this, the Fed will buy \$400 billion of long-term Treasury bonds by next June, while selling an equivalent amount of shorter-term securities. The plan is designed to lower long-term interest rates, while keeping short-term rates little changed.

Nevertheless, as interest rates have already been cut to record lows, we doubt this new measure of monetary stimulus will be effective to restore consumer confidence and boost GDP growth. Although our base case scenario does not assume that the US economy will fall back into recession next year, the downside risk is growing, particularly given the persistence of elevated unemployment, weakness in the housing market and deeper contagion from the European sovereign debt crisis. As a result, US GDP growth is expected to slow further to 1.5% in 2012 (compared with 2.0% in our previous report) before economic conditions slowly improve from 2013 onwards.

Our GDP forecast for the **EU-27** has been revised lower since the previous forecast, as growth in Germany and France, the two largest economies in the Eurozone, both stalled in the second quarter. In our view, the sharp deceleration in Europe's major economies will not only hit the Eurozone's overall growth this year

Annual World Real GDP Growth Rates



Source: Thomson Reuters GFMS; IMF



Industrial Production				
(%)	2011	2012	2013	2014
Base Case Scenario	2.3	1.0	1.8	1.4
(Probability 45%)				
Scenario B	2.6	2.0	3.0	2.5
(Probability 40%)				
Scenario C	1.5	0.8	1.5	1.2
(Probability 15%)				
Source: Thomson Reuters GFMS, IMF				

but also threatens to complicate the challenge facing the region's policy makers as they seek to combat the escalating debt crisis. Given the excessive level of public debt in many European countries, we would expect further austerity measures to take place in a number of southern European countries over the next two years, which in turn will continue to drag down the region's growth. Real GDP growth in 2012 is therefore forecast to fall to 1.0% with a possibility of contraction in some of the quarters before a gradual recovery takes place in the final two years of the forecast.

The nascent recovery in **Japan** was thrown off course by the twin disasters of the earthquake and the tsunami and we are now expecting a contraction of 0.5% in 2011. While manufacturers have now largely overcome post-quake production disruptions and the threat of power shortages, the outlook for the country's industrial production remains murky. To a large extent, this is due to a major slowdown in western economies coupled with a stronger Japanese yen, which could put considerable pressure on the export sector. Over the next three years, given more pipeline increases together with further construction in the earthquake-damaged regions, the Japanese economy is expected to grow at an average of 2.0% per annum.

China's economic growth slowed further in the second quarter after the government stepped up its battle against surging inflation. Although CPI eases slightly to 6.1% in September, the government will still treat managing inflation as the priority in the near term. Nevertheless, China is unlikely to see a hard landing, as the economy should benefit from continued policy efforts to raise disposable incomes, with the drivers of growth shifting increasingly from the public to the private sector. We therefore forecast China's economy to expand by at least 9.0% this year. Over the next two years, despite a continuation of a tighter monetary policy, growth is likely to remain solid and stable, albeit at a slightly slower pace of 8.5% in 2012 and 2013. Thereafter, growth is expected to moderate, albeit at a still healthy 8.0% in the final year of the forecast.

Elsewhere in the developing world, GDP growth in 2011 and 2012 is forecast to moderate amid the world's sovereign debt crisis and, as a result, faltering global demand. Thereafter, we expect growth rates to return closer to trend levels over the last two years of our forecast. As with China, the slowdown is largely a result of a continued tightening of monetary and fiscal policy in many countries where inflation has already picked up rapidly since the latter part of 2010. Nevertheless, economic growth in emerging markets will remain at far higher levels than in the developed world, helped by growing manufacturing capability and expanding domestic spending. **Indian** growth for example is projected to average 7.7% in 2011-14. **Brazil** and **Russia**, the other two BRIC nations, are expected to sustain average growth rates of 3.7% and 3.8% respectively from 2011 to 2014.

Despite the US credit downgrade, the **dollar** has gained marked strength, particularly against the euro, over the last couple of months, as the \$/€ fell to a nine-month low by early October. This, however, was more due to international euro weakness, after mounting concerns over the Greek debt crisis saw investor flight to safe haven assets such as the US treasury bonds and the dollar. Furthermore, the increasing investor appetite for the greenback should also be considered in the context of limited options left for investors, as the other two popular international safe-haven currencies, namely the Japanese yen and the Swiss franc came under fire (due to central banks' intervention in the currency market). Under our Base Case, we forecast the US dollar appreciation against most currencies from 2012 onwards. This move will be further assisted by the fact that by then the Federal Reserve will, finally, start to raise interest rates. We therefore expect the US dollar to average \$1.25/€ by 2014.

2.4 Scenario B

Scenario B, our most bullish view of the world economy, has global GDP growth at an average of 3.4% from 2011 to 2014. Critical to this position is a very

US CPI				
(annual averages, %)	2011	2012	2013	2014
Base Case Scenario	2.8	3.1	3.9	3.4
(Probability 45%)				
Scenario B	3.3	3.9	5.5	5.8
(Probability 40%)				
Scenario C	2.4	2.1	2.0	2.0
(Probability 15%)				
Source: Thomson Reuters GFMS, IMF				



slow or in some cases non-existent withdrawal of fiscal stimuli and further quantitative easing.

Against this hugely accommodative backdrop, it is unsurprising that many countries experience rising inflation, especially the **United States**, albeit from a very low base. This, combined with a rapid recovery in confidence regarding the world economy, could encourage investors to use the dollar to fund carry trades, which in turn will result in a rise in the broader commodity complex. This leads us to forecast an average inflation rate in the United States of 3.9% in 2012, rising to 5.8% in 2014. US GDP growth is forecast to average 2.3% from 2011 to 2014. Even given the reluctance to tighten monetary and fiscal policy under this scenario, we believe it would become imperative for the United States to take measures towards the end of the forecast. Central to this will be mounting concerns over the sustainability of US long-term debt, as we would argue that other rating agencies will also have downgraded the US credit rating by that time.

Under Scenario B, growth in **Europe** and **Japan** is stronger than under our Base Case over the forecast period, although to a lesser degree than in the United States. Correspondingly, GDP growth will measure 1.6% for the EU-27 in 2011 before slowing slightly in 2012. In Japan, the economic consequences of the earthquake will be smaller than under the Base Case, as economic growth slows to 1.0% in 2011 before a fair recovery to 2.5% in 2012.

Developing countries' export sectors fare better from the stronger GDP growth in the developed world forecast under this scenario. This is particularly the case for **China**, which under Scenario B experiences an average annual growth rate of 9.2% from 2011 to 2014, despite a minor slowdown over the period.

As regards the **US dollar**, the inflationary environment and lack of withdrawal of monetary stimulus will see a weakening of the American currency, at least on an annual average basis. Accordingly the euro is projected to strengthen from \$1.45/€ in 2011 to average \$1.49/€ in 2012-2014. (Implicit in this scenario is that the European sovereign debt crisis does not deteriorate sufficiently to undermine the single currency.)

2.5 Scenario C

The drag from balance sheet rebuilding pushes down GDP growth rates under **Scenario C**. Accordingly, many advanced economies border on very low levels

of inflation. Therefore, global GDP growth is a poor 2.0% this year, with developed economies massively underperforming even that meagre result. This is followed by marginally slower growth in 2012. Thereafter, growth rates improve but remain at historically low levels, especially for a post-recession phase of economic growth.

A key factor behind these sluggish growth rates is the high level of indebtedness of governments in developed economies, including the world's largest single economy, the **United States**, where, as a result, fiscal tightening also eventually takes place. As a consequence, that country sees just 1.2% growth in 2011 as the economy stagnates and the ongoing loose monetary policy is ineffective in ensuring that economic recovery continues at the former pace. This would be in large measure due to stubbornly high unemployment rates, consumers' still massive debt burdens and, at best, a stagnant housing market. Furthermore, exporters in the United States are set to struggle under this scenario as the US dollar appreciates to an average of \$1.27/€ in 2012 and to \$1.21/€ in 2013. As a result, US GDP is set to average 1.1% from 2012 to 2014.

In **Europe**, the outlook also looks bleak as economic growth is restrained by the need to limit the increase in government debt and by high levels of unemployment that dampen consumer spending. In fact, both the spate of austerity measures in this region plus an array of further steps to address the fiscal deficits could lead to EU-27 GDP increasing by only 1.0% in 2011. Thereafter, we forecast EU-27 GDP to average 0.9% between 2012-2014.

Given the difficulties faced by the **Japanese** economy over the past 20 years and the recent disaster, it is unsurprising that it struggles to avoid an outright return to recession under Scenario C. Therefore, GDP is forecast to contract by 1.1% in 2011, before there is a very modest improvement in the GDP growth rate in the final years of the forecast.

Given the industrialisation taking place in many developing countries they will continue to be the best performers. Needless to say **Chinese** growth will be the strongest among the larger countries in this group, although by its standards an increase of 8.1% this year would represent a fair dip. **India's** growth rate is also expected to hold up well, with an average rate of 6.6% from 2011 to 2014.



3. Mine Production

3.1 Introduction

Global platinum supply is forecast to rise by 0.2M ounces or 3% this year. By 2014 global mine supply of platinum is forecast to be almost 12% higher than it was in 2010, though marginally falling short of 2006's record volume. The most meaningful change this year is expected to come from Canada's base metal mines, with the country's overall production expected to double year-on-year (and remain around these levels for the rest of the forecast). Growth is anticipated in Zimbabwe this year, and expected to continue to a record level of 0.4M ounces by 2014 as recent project developments ramp up. Although not expected to be a major feature in terms of change in 2011, the world's largest producer of platinum, South Africa, will remain a core area of global supply growth in subsequent years, unless metal prices continue to decline, forcing closure across multiple shafts. Widespread price-driven closure is not expected in our forecast on the basis of the prices presented in Chapter 1.

Annual palladium output from mines is expected to increase by 4% in 2011 to a total of 6.9M ounces. Thereafter production will increase successively each year to 2014 by which time output is expected to have reached 7.3M ounces, 0.7M ounces, or 11% of additional supply relative to production in 2010. As has been the case for platinum, renewed mining activities in Canada have already grown, and will continue to increase output through 2011, representing the bulk of anticipated production increase in the near term, along with modest gains in Zimbabwe and the United States, with South African and Russian output expected to be relatively flat. Further out, pent-up development growth from established South African mines and

greenfields projects is expected, adding over 0.2M ounces between 2011 and 2014, as well as growth in Canada, and to a lesser extent in Zimbabwe and the US.

Given the extreme market turbulence and impact on metal prices recently it is worth noting that although we make 'contingency allowance' to attempt to account for unforeseen disruption to mining (of just over 0.2M ounces of platinum per year on an ongoing basis), we do not currently expect lower prices to have a widespread impact on production (shaft closure) in the forecast period. Although between the end of August and mid-October the South African 4E basket has dropped by almost 17% (in dollar terms) the weakening of the dollar:rand exchange rate has fallen from 7 to around 8 rand to the dollar, which have sustained the drop in rand-based pricing to a more modest 7%, to around R9,830/oz (4E). On the basis of analysis of total cash costs over 95% of the South African industry is currently producing at costs below this level. Behaviour of 2008/9 shows that producers are highly resistant to making cuts to production (and therefore labour) that will be opposed by the government and unions, and be slow and costly to reverse if necessary. Furthermore, production cutbacks contravene the current producer strategy of reducing unit costs through increasing production. It should also be borne in mind that a number of the highest cost operations are joint ventures between the large mining houses and black economic empowerment (BEE) partners, and such mines may well be insulated against closure in order for mining companies to remain on track to comply with BEE divestment targets.

Platinum Mine Production					Palladium Mine Production				
(000 ounces)	2011	2012	2013	2014	(000 ounces)	2011	2012	2013	2014
South Africa	4,717	4,908	4,980	5,144	South Africa	2,651	2,787	2,818	2,892
Russia	811	790	782	796	Russia	2,715	2,668	2,647	2,661
Canada	293	265	268	286	Canada	556	538	593	738
United States	120	129	129	129	United States	399	419	421	421
Zimbabwe	337	332	373	418	Zimbabwe	258	253	282	316
Others	119	122	144	154	Others	285	292	307	314
Total	6,397	6,546	6,676	6,927	Total	6,864	6,957	7,067	7,343
y-o-y %	3%	2%	2%	4%	y-o-y %	5%	2%	3%	2%
Source: Thomson Reuters GFMS					Source: Thomson Reuters GFMS				



South Africa

XRefined output of platinum and palladium is expected to be flat in 2011, owing to labour disruption and several technical operating difficulties. After this hiatus we do expect output to return to growth, with modest consecutive production gains for both metals out to 2014. By 2014 platinum production from South Africa is forecast at 5.1M ounces (an increase of 8% from 2010) with palladium totalling 2.9M ounces (up 9% from 2010). These figures include allowances made for under-delivery from possible industrial action, enforced mine suspensions and possible power outages.

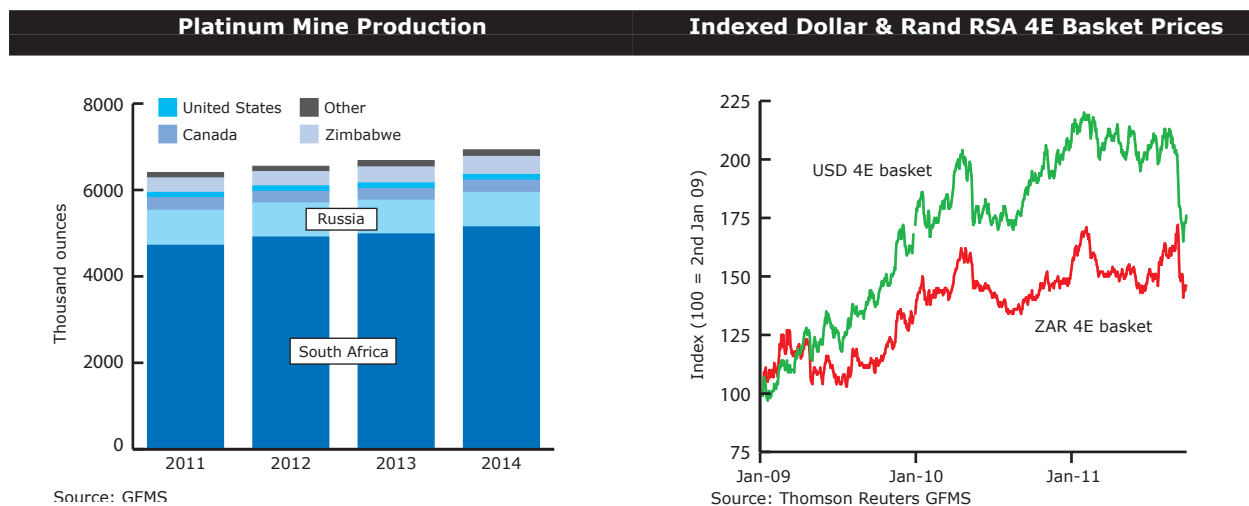
In the last forecast we highlighted the rising incidence of safety problems impacting available production days. Associated with that issue we look to labour productivity which has lagged our expectation of improvement in this area. Although the frequency of safety stoppages has lessened somewhat in recent months, we have yet to see evidence of substantially improved productivity. As improvements in both production and the relative control of unit costs are contingent on improved productivity, our forecast depends on producers achieving efficiencies close to their targeted improvements, especially in light of the dramatic price correction that occurred in September.

Another area of concern has been the possible impact of strike action in the season of wage negotiations, which for many platinum producers are a biennial event. Several producers have now reached agreements with the unions including, importantly, Anglo American Platinum (Amplats) and Impala Platinum (Implats), as well as Royal Bafokeng Platinum (RB Plats) and Anooraq Resources. In most cases the settlements entail increases of between 8.5% and 10% in basic salaries over the next two years. Although Lonmin and Northam's negotiations are ongoing, the

risk for the industry to grind to a halt this year appears to have passed, given the settlements by companies that represent the bulk of the production base.

We maintain a rising production profile for Amplats' refined production, reaching 2.75M ounces by 2013 and rising above 2.8M ounces by 2014. Built into this projection is an assumption, in line with company strategy, of higher output from Rustenburg: this will be aided by the recently restarted Khuseleka 2 shaft, phased replacement at Bathopele and some growth from the Thembelani 2 shaft project. Overall, the company is aiming to return annual production from its Rustenburg units back to 0.7M ounces in the medium term. In the long term, the company has alluded to a strategy that will utilise existing Merensky infrastructure to more cost effectively access UG2 ore rather than heavily investing in sinking new, deep-level shafts, such as those provisionally planned as Tumela 4 and Union 'Deep shaft'. Amandelbult Dishaba has yet to deliver on its East Upper expansion but after a softer than expected couple of years is now scheduled to do so over the next three years (post-2011). Another key area of growth (and production flexibility) is at Mogalakwena, where, by 2014, an expansion of the North Concentrator to accommodate increased mining rates is expected to be well underway.

Representing a critical part of the strategy common amongst the majors of driving down unit costs by expanding metal production from core assets, further growth is expected at the Impala Lease Area and Lonmin's Marikana complex. At the Impala Lease a build-up from 0.95M ounces this year to the 1.00M ounce level is expected to be achieved by 2015, as 20 shaft and 16 shaft come online and raise the mine's Merensky ore volumes. This remains our expectation in spite of recent guidance that development at 20





shaft is experiencing complications and accordingly will fall short of earlier anticipated volumes in its ramp up phase. Lonmin's Marikana is expected to continue to ramp up volumes at Hossy & Saffy and later deliver output from the high grade Merensky reef to be exploited via the K4 shaft, currently under development. Central to the company's strong growth profile (to produce 0.95M ounces by 2015) is to achieve substantial productivity improvements across the Marikana operations. We are cautiously optimistic that this can be achieved but we apply a 'haircut' to this projection and forecast a moderated 0.9M ounces at full production (including additional build-up at Pandora). We do not currently make allowance for output from Limpopo (following the recently announced transaction with Shanduka and the possible redevelopment of the asset) within our forecast.

Turning to the smaller producers (in the context of platinum), Xstrata and Eastern Platinum both experienced a difficult first half at their respective Western Limb operations, Eland and the Crocodile River mine (CRM). At Eland challenging underground conditions coupled with declining open cast mining has been behind its shortfall, while at CRM much of the recent production shortfall and associated unit cost inflation is tied to labour unrest and lost production days. It should be noted though that these are transient issues that therefore in due course pave the way for production increases at both operations. We assume improvements will start to be realised by the end of this year, to be subsequently further improved as ore volumes from the Crocette decline ramp up. Eland is forging ahead with development of two declines with substantial production expected from the Western Decline by 2013 and commissioning production from the Eastern Decline. Collectively we forecast a rise of 0.20M ounces of additional platinum from these mines by 2014.

Looking to the prospects from greenfields projects, we expect Northam's Booyesendal and Eastplats' Mareesburg open pit to be ramping up production by 2014 and together we forecast they will contribute almost 0.1M ounces of additional platinum supply that year. These projects are well funded and ahead of others in terms of allocation of grid power and water permits. Those miners that have not already secured grid power allocation will struggle to do so before 2013 at the earliest in our opinion, especially considering the recent guidance by Eskom that the first unit of its Medupi power station (the parastatal utility's first major new build) is facing delays.

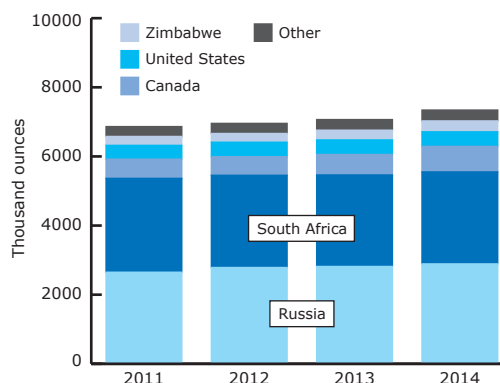
Russia

In Russia we expect the production rate of PGMs to remain relatively stable to 2014. Central to this assumption is rising throughput to counter a declining grade profile at Norilsk Nickel's operations.

As background information, a research trip to Kola and Norilsk in the summer highlighted a number of potential developments underway at the company's domestic assets. Highly capital intensive moves to improve the environmental footprint and efficiency of the downstream base metals processing circuit for Norilsk Nickel's domestic operations appear to be gaining traction. This is focussed on a transition away from, where viable, a largely pyrometallurgical process route to a 'roast-leach' beneficiation process that will improve base metal process recoveries and productivity efficiencies. Although the benefits of this will largely be confined to the base metals, the company also highlighted plans to modernise the circuits in Norilsk's 'Metallurgical Shop Number 2' (the high grade PGM concentrator) to replace a relatively labour-intensive batch process that involves additional smelting and electrolytic technology with, it is assumed, more automated procedures. As far as we could ascertain, maintaining metal production volumes seems credible in spite of a move to increasingly high proportions of lower grade disseminated ore. This will in part be accommodated by a major concentrator expansion at Talnakh, through the addition of a new 10 Mtpa milling circuit scheduled for construction in 2012.

Underlying these changes at Russia's main PGM producing mines, alluvial platinum production, primarily from Russia's Far East, is expected to continue a gradual decline over the next three years. Anecdotal information suggests that output from the alluvial sector could actually buck the trend and recover

Palladium Mine Production



Source: Thomson Reuters GFMS



overall production levels somewhat in coming years, although we have yet to corroborate this and do not yet factor additional volumes into our forecast. Should improvements in this sector be realised it would only affect platinum production, as contained palladium in Russia's alluvial PGM deposits is minimal.

North America

Premised on a recovery from a challenging year in 2010 at the Stillwater mine, in 2011 we expect PGM output from the United States to push back above 0.50M ounces (combined platinum and palladium) in line with Stillwater's guidance, which was raised by 3% at the last quarterly results call. Modest additional gains are then expected, with output to reach 0.13M and 0.42M ounces of platinum and palladium respectively for the remainder of the forecast. High grade material on the 'East Side' area of the Stillwater mine features in the near term mine plan and, although ground conditions in this area are expected to be challenging, the recently lifted production target suggests to us development may be more manageable than initially planned for. At East Boulder there is also scope for increased output from current volumes. Potential long term expansion opportunities at both mines are also under study but we are of the view that they are not sufficiently advanced to alter the production profile by 2014.

In Canada, Vale's Sudbury (nickel) mines and process facilities have been operating at notably strong levels of production (from the perspective of PGMs) through the year so far. Coupled with a full year of production at North American Palladium's Lac des Iles, Canadian output is expected to or even surpass 'normal' levels this year with palladium output set to recover back above 0.5M ounces and platinum to set a new record production total of almost 0.3M ounces. In the case of the platinum, the emergence and rapid ramp-up of Nickel Rim South last year has been significant and will continue strongly this year. Information underlying Sudbury's solid performance has not been publicised but we expect output to settle back to lower levels in the medium term.

Other regions

Although the news is now six months old, it is worth reiterating Zimbabwe's indigenisation policy, announced in March, which stated the requirement 'for foreign companies to dispose of a minimum 51% of shares or interests to designated indigenous Zimbabwean entities within six months. The value of shares or interests to be disposed of are to be agreed between the minister and the non-indigenous mining business concerned,

"which shall take into account the State's sovereign ownership of the mineral or minerals exploited or proposed to be exploited by the non-indigenous mining business concerned."

Producers' proposals for the indigenisation divestments centred around an earlier established model of a 26% equity divestment with the remainder of the 51% figure to be made up by previously relinquished ground and social and infrastructure spending. These proposals were duly rejected by the government. Our expectation is for a sweetening of the previous offer by the producers rather than anything resembling a full 51% divestment. That said, we believe that the potential production impact over a three year horizon would be limited even if the government's plans were maintained; expansion and construction activities at several operations are well underway and these production gains will be realised. Unki's initial production performance has been stronger than anticipated thanks to output from accumulated stockpiles and the Ngezi Phase 2 expansion continues. Despite the heightened political risk, we expect production from Zimbabwe's mines to have surpassed 0.4M ounces and 0.3M ounces of platinum and palladium respectively by 2014.



4. Autocatalyst Scrap

4.1 Introduction

Last year, the recovery of PGMs resumed its upward growth path, following the losses posted in 2009, the first ever full year of declines seen in the industry. 2010s improved total owed much to the healthier price trend, together with the impact of the various 2009 "cash for clunkers" schemes, which saw some vehicles scrapped during early 2010. Last year also saw a partial destocking of vehicles, which had been accumulated by breakage and scrap yards, encouraged by healthier PGM prices, although this trend appeared to be more common in the United States.

the average vehicle life used to model the scrap series) and how offtake varied at this time.

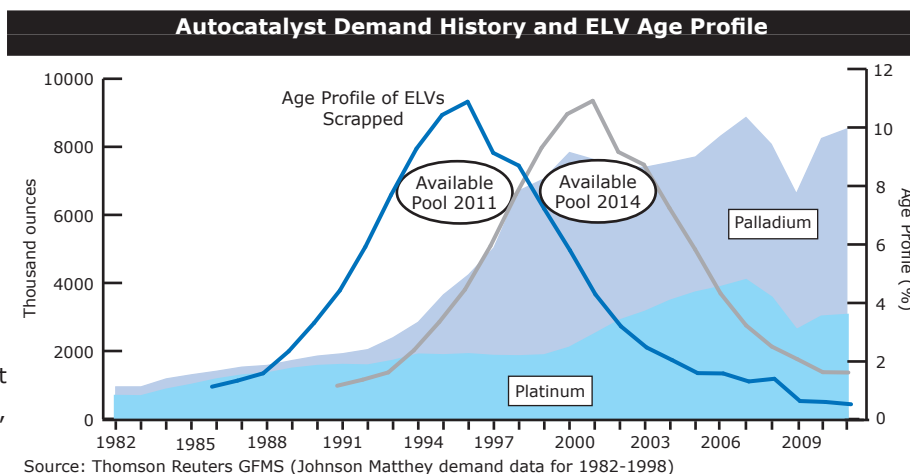
The key reason for the revision is the impact of the stalled economic recovery, especially in the United States (and how this has resonated in the real economy) which appears to have encouraged consumers to delay replacing their existing models. This in turn has resulted in a slight increase in the average age of vehicles being scrapped this year, with a bias more towards the 14-year mark, rather than those vehicles that are nearer 12 years old (in

4.2 Outlook

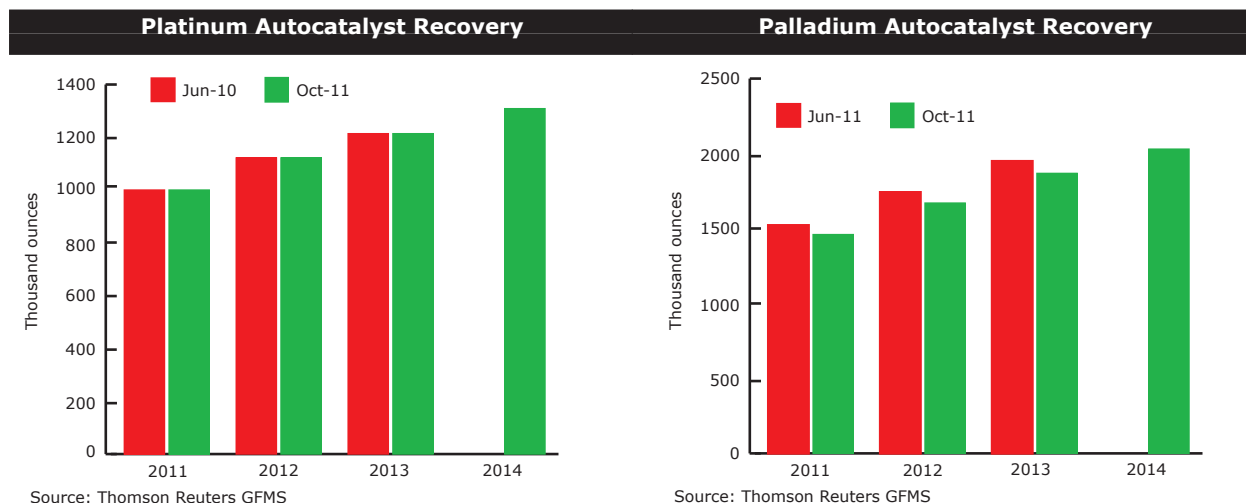
Looking ahead, 2011-14 is expected to see a period of uninterrupted gains in the recovery of platinum and palladium from spent autocatalysts. Following estimated growth this year of 10% for platinum (to a total of 986,000 ounces) and a 13% lift for palladium (to 1.5M ounces), by 2014 the respective totals are forecast to rise to 1.3M ounces and 2.0M ounces.

Amendments since the June report are modest and are confined to the palladium series, which has been revised lower, with the average 2011-13 growth rate now at 13%, compared with our previous expectations of a 15% improvement over this time frame. The fact that the platinum data series has been left unchanged is related to the historical level of autocatalyst demand (we focus on demand around 12-14 years ago as this is

terms of the average). Given the relative stability of platinum demand during the late 1990s this change has had little bearing on the recovery of this metal from scrapped autocatalysts. In fact any potential downward revision to the platinum scrap series has been offset by the impact far stronger platinum prices (through to September that is) have had on platinum autocatalyst scrap supply. However, the late 1990s did see a steep rise in the use of palladium (especially in the United States and Europe) and so focusing on earlier-dated vehicles (even marginally so) has involved looking at



Platinum Autocatalyst Recovery					Palladium Autocatalyst Recovery				
(000 ounces)					(000 ounces)				
	2011	2012	2013	2014		2011	2012	2013	2014
North America	467	510	513	511	North America	937	1,048	1,133	1,190
Europe	343	391	447	513	Europe	330	391	449	501
Japan	72	80	93	100	Japan	103	116	135	147
Other regions	104	126	145	166	Other regions	89	115	152	192
Total	986	1,106	1,197	1,290	Total	1,460	1,670	1,868	2,030
y-o-y	10%	12%	8%	8%	y-o-y	13%	14%	12%	9%
Source: Thomson Reuters GFMS					Source: Thomson Reuters GFMS				



models with less palladium-rich catalysts, hence the downward revision to the forecast series.

Returning to the trends we expect to see out to 2014, the growth in **platinum** scrap supply is expected to gradually slow over the next three years. This will be largely due to developments in North American recycling (which has accounted for the largest share of platinum scrap), registering virtually no growth during the final two forecast years (as well as failing to recapture the record levels achieved during the late 2000s), a trend which will be partially offset by still robust growth in Europe. As a result, it is feasible that, by 2014, European supplies, posting successive record highs, will match the level of autocatalyst supply emerging from North America.

The contrasting performances is due to the respective differences in platinum offtake at the turn of the millennium. While demand in the far more mature US market (where catalysts were first deployed in the 1970s) was trending lower at this time, Europe was benefiting from healthier platinum demand, itself due to the increasing popularity of diesel vehicles (from the mid-1990s). One mitigating factor, impacting the European recycling total, is the significant export of second hand vehicles, principally from Germany, destined in particular for the Middle East and Africa. This long-established trade was briefly curtailed during 2009 as government scrappage schemes required qualifying models to be recycled domestically (Germany's scheme was by far the largest government-backed programme in Europe). Although early 2010 still saw an important share of these units being recycled (especially in Germany), last year also witnessed the resumption of the export trade, which reduced the available pool of vehicles that could have been scrapped in Europe.

Elsewhere, the recovery of platinum from Japan will grow over the forecast period, but at a far slower pace than in recent years, reflecting the decline in platinum offtake over the 1990s. In contrast "other regions" is set to achieve a far healthier rate of growth. Markets such as South Korea have benefited from the roll-out of emissions legislation, although it is only recently that this has required the use of PGMs, which explains why only in 2013-14 will the recovery of platinum improve noticeably.

This development also explains the growth in "other regions" **palladium** autocatalyst scrap supply during this period, which will comfortably eclipse supplies emerging from Japan by 2014. However, North America and, to a lesser extent, Europe will continue to dominate the global recovery of palladium from spent autocatalysts. Taking each in turn, the rate of growth in North American scrap is expected to ease noticeably as we approach 2014, from the forecast double-digit percentage gains predicted for 2011-12. This relates to the marked slowdown in North American (autocatalyst) palladium offtake in 1999 and 2000 (the timeframe which accounts for the bulk of the scrapped models by 2014), which gave way to an outright decline in usage during the early 2000s. Even so, North American recycling will still post the largest gains over the coming three years (comparing its expected contribution in 2014 with the 2011 total).

European autocatalyst demand did not escape the impact of sharply higher palladium prices in 2000-01, although the rate of attrition in the early 2000s was far less pronounced than in North America. As a result, Europe recycling is still slated to achieve double-digit percentage gains over the entire 2012-14 forecast, which in turn will help push the global total for the first time through 2.0M ounces by 2014.



5. Above-ground Stocks

5.1 Introduction

This analysis includes our estimate of stocks held in the terminal market (implied from historical trade data), on futures exchanges and in physically backed exchange traded funds (ETFs). It also includes estimates for excess Russian government stocks of palladium (that which is deemed surplus and scheduled for sale), based on the assumption that sales of this palladium will cease during the term of this forecast. In the absence of formally reported data regarding above-ground stocks in the terminal market (the largest element of our estimates) and uncertainties over Russian state holdings, we reiterate that these assessments represent our best estimates, and thus should be viewed as broadly indicative rather than definitive. This analysis excludes any stocks held separately by producers and consumers (industry stocks) although movements in such holdings can be inferred from time-to-time. As usual, alterations to our forecast supply/demand balances flow through to changes in the outlook for above-ground stocks.

While there remains a greater abundance of palladium stocks relative to platinum, we expect a significant degree of convergence over the forecast period. Platinum stocks have been rising over recent years (notably so since 2009) and are expected to grow still further over the forecast period as a significant gross surplus continues to occur. In the case of

palladium, after a slender gross deficit in 2009, a more substantial shortfall emerged in 2010 and is expected to be followed by further gross deficits (although lower on average than last year's) through to 2014. Nonetheless, this will result in a commensurate decline in global above-ground palladium stocks.

By the end of 2011, we estimate that above-ground stocks of platinum and palladium at 4.5M ounces and 10.9M ounces respectively (including ETF allocations). This represents a palladium to platinum ratio of 2.4 to one, with the ratio having contracted from four to one since the end of 2009. By the end of the forecast period, we expect this ratio to contract further to only 1.4 to one in favour of palladium, with stocks at the end of 2014 projected at 6.5M ounces of platinum and 9.4M ounces of palladium.

5.2 Platinum

Our supply/demand estimates indicate a persistent gross surplus of platinum over the last seven years (including 2011), increasing in scale and approaching 0.9M ounces in both 2009 and 2010, with the excess reducing somewhat in 2011 to an estimated 0.71M ounces. This represents a cumulative gross surplus since 2005 of almost 4M ounces, which has driven an equivalent increase in global above-ground platinum stocks. Furthermore, in 2009 this was compounded by

Changes to Platinum Above-Ground Stocks - Zurich and Futures Exchanges Combined

(M ounces)	2009	2010	2011	2012	2013	2014
Closing level at end-2008	1.39	-	-	-	-	-
Implied industry stock sales (net)	0.67	-	-	-	-	-
Gross surplus/(deficit)	0.88	0.88	0.71	0.88	0.57	0.56
Implied year-end stocks (minimum)	2.93	3.82	4.52	5.40	5.97	6.52
Fabrication demand	6.41	6.81	7.24	7.33	7.83	8.18
Stock cover, months	5.5	6.7	7.5	8.8	9.1	9.6
ETF holdings	0.68	1.25	1.55	1.70	1.50	1.40
Stock liquidity (maximum)	77%	67%	66%	68%	75%	78%
Stock cover, months (liquid stocks)	4.2	4.5	4.9	6.1	6.8	7.5
Source: Thomson Reuters GFMS						



significant industry de-stocking (some 0.7M ounces) at the depths of an exceptionally severe recession. As a result, above-ground stocks of platinum have risen considerably over recent years, ballooning since 2009 when fabrication demand also fell heavily. This extended period of underlying gross surplus contrasts starkly with the several years of deficit which preceded it. While that era of deficits is now a distant memory (2004 was the last of those years), it had caused a progressive decline in global platinum stocks, so much so that by the time it finished stocks were perilously low at less than 1M ounces.

Looking forward, we estimate a gross surplus of platinum averaging 0.8M ounces in both 2011 and 2012 (somewhat lower but not dissimilar to the outcomes in 2009 and 2010), but expect this to ease in 2013 and 2014 to averages of a little under 0.6M ounces. In total, this represents a cumulative gross surplus over the 2011 to 2014 period of 2.7M ounces of platinum. This will flow directly into above-ground stocks, inflating them to a total of 6.5M ounces by the end of 2014. In terms of demand cover, this enlarges over the same period to the equivalent of almost ten months cover (basis fabrication demand in 2014) in contrast with just over seven months cover at the end of 2010. This is despite an increase in platinum fabrication demand (2014 versus 2011) of almost 1M ounces.

These estimates of platinum stocks include allocations (historical and forecast) to the currently listed physically-backed ETFs. Whilst these allocations are integral to global above-ground stocks, platinum held in these products is assumed to be unavailable for lending or other trading purposes and remains separate from the pool of liquidity in the terminal market. At the end of 2011, we estimate allocations across all ETFs totalled 1.55M ounces of platinum, representing an approximate one-third share of overall above-ground platinum stocks. However, over the forecast period, ETF holdings are expected to ease somewhat, with light disinvestment emerging from 2013. This means that, with above-ground platinum stocks trending higher, ETFs' share will retreat over the forecast period, thus reducing the extent to which they serve as a drain on physical liquidity. If ETF allocations are excluded, demand cover afforded by the remaining stocks of platinum is projected at just under five months at the end of 2011, a measure which is expected to lengthen to more than seven months by the end of 2014.

As a cautionary note, we highlight that visible stocks of platinum (including ETF allocations) are held in the form

of ingot/plate, whereas the bulk of fabrication demand consumes platinum in the form of sponge; a form which can be readily dissolved into pre-cursors for catalyst applications (including autocatalysts). Aside from metallurgical applications such as jewellery and glass, where ingot/plate is the required form, should a major disruption in new supply require the mobilisation of above-ground platinum stocks, this would necessitate the conversion of ingot/plate into sponge. This would be subject to cost and time considerations, as well as the availability of processing capacity.

5.3 Palladium

While there have been numerous claims of an imminent end to sales of palladium from Russian government inventories, we think it may be a while yet before the curtain is finally lowered on these disposals. Anecdotal information suggests that sales of around 0.8M ounces occurred in 2010 (a view supported in part by trade data) and will most likely continue at similar levels in 2011 as well. However, we think these are likely to be the last years in which elevated sales occur and that the end-game will then be played out with residual sales (declining successively) in 2012 and 2013. This view is based on anecdotal information from sources we believe to be reliable (although this cannot be guaranteed) and is further supported by our own modelling of the historical profile of Russia's palladium stocks.

Given the strong probability that inventory sales from Russia will cease within the horizon of this forecast, we have elected to consolidate an estimate of these holdings into our long-standing assessment of above-ground palladium stocks. (This estimate is entirely our own – Russian government palladium stocks remain a state-secret.) Despite the uncertainties this entails, we believe there is a greater benefit in attempting to provide as complete a picture as possible, even though revisions may be necessary from time-to-time. With this included, we estimate global above-ground palladium stocks of approximately 11M ounces by the end of 2011. As a result of repeated gross deficits, global palladium stocks have been eroded over the last five years (including 2011 estimates) by a total of some 2M ounces – a process we expect to continue over the forecast period as further gross deficits recur. With the total of these deficits amounting to over 1.5M ounces, global palladium stocks are expected to fall to a little over 9M ounces by the end of 2014 – a cut of some 14% over the next three years. Relative to demand, our stock estimate for the end of 2011 represents 15 months of cover. However, with stocks declining and



fabrication demand reaching 10M ounces in 2014, cover by the end of the forecast period is expected to contract quite sharply to the equivalent of around eleven months demand.

As with platinum, these estimates include allocations to palladium's suite of ETFs, which at the end of 2011 we estimate will total slightly less than 2M ounces; some 18% of overall palladium stocks. By the end of 2014, palladium holdings in ETFs are forecast at broadly similar levels, thus representing a static block relative to overall stocks which we expect to trend lower. If ETF holdings are excluded, remaining stocks afford a still comfortable twelve months of demand cover for palladium this year. However, by the end of 2014 this is forecast to shrink to around nine months cover basis fabrication demand at that time.

As with platinum, it should be noted that visible stocks of palladium are held in the form of ingot/plate, while the bulk of fabrication demand requires metal in the form of sponge. In fact, with its less developed role in jewellery applications, the dependence on palladium in sponge form is even greater. Consequently, in the event of a major disruption in new supply, the stock levels we forecast in this report may be less comforting than they initially appear as their mobilisation would be constrained by the process of converting ingot/plate into sponge and the availability of capacity.

Changes to Palladium Above-Ground Stocks - Zurich and Futures Exchanges Combined

(M ounces)	2009	2010	2011	2012	2013	2014
Closing level at end 2008 *	11.89					
Gross surplus/(deficit)	(0.09)	(0.58)	(0.34)	(0.58)	(0.50)	(0.44)
Implied year-end stocks *	11.80	11.22	10.87	10.30	9.80	9.36
Fabrication demand	7.60	8.65	8.84	9.39	9.62	10.00
Stock cover, months	18.6	15.6	14.8	13.2	12.2	11.2
ETF holdings	1.17	2.26	1.96	2.16	2.26	1.96
Stock liquidity (maximum)	90%	80%	82%	79%	77%	79%
Stock cover, months (liquid stocks)	16.8	12.4	12.1	10.4	9.4	8.9

*Includes estimated Russian government holdings as of March 2011 publication.

Source: Thomson Reuters GFMS



6. Fabrication Demand

6.1 Introduction

This chapter covers platinum and palladium demand in autocatalysts, jewellery and their respective industrial uses. The final element of demand, retail investment, is reviewed in Chapter 7. We show our fabrication series in two formats. Industrial uses are featured on a net basis, as the scrap generated in their 'closed loops' is price inelastic and relatively stable; it is their demand for fresh metal that is crucial. In contrast, the 'open loops' of jewellery and autocatalyst offtake are shown on a gross basis, with a separate scrap series, since recycling here can be volatile and price-dependent.

Platinum fabrication is forecast to see annual growth averaging 5.3% for 2012-2014, taking offtake in the final year to 8.1M ounces, a record basis our series going back to 1999. Some of this rise is down to improving autocatalyst demand as European diesel car sales recover. However, substitution in diesel means platinum's 2011-14 rise is less than half palladium's absolute gain in this sector, and the 2011-14 rise for platinum jewellery should also exceed autocatalysts' gains. Jewellery's rise (overwhelmingly in China) is largely due to periods of a notable platinum:gold price discount, the retreat in platinum prices from 2013 and ongoing GDP gains. Changes for the other demand segments over 2012-14 are small if combined, but that disguises marked gains for our other industrial category and losses for glass and petroleum in particular. Changes since the last report chiefly comprise a cut to European autocatalyst demand in 2012 and 2013 due to economic problems and gains for glass in all years bar 2012.

Three years of fabrication growth is also expected for palladium, if at a slightly slower average pace between 2012-14, of 4.4% a year. This takes the total by 2014 to 9.9M ounces, also an all-time high basis our records. The increase from 2011 to 2014 is overwhelmingly due to the 1.3M ounce rise in autocatalyst demand, with all regions, and in particular China, enjoying gains of note thanks to rising car sales and substitution from platinum. A key area moving in the opposite direction is jewellery, where falling sales in China bring about a 21% drop in the total when comparing 2014 with this year. The main revisions since the June report are cuts in all years due to a gloomier economic outlook for both electronics offtake and autocatalyst demand, mainly in Europe and North America.

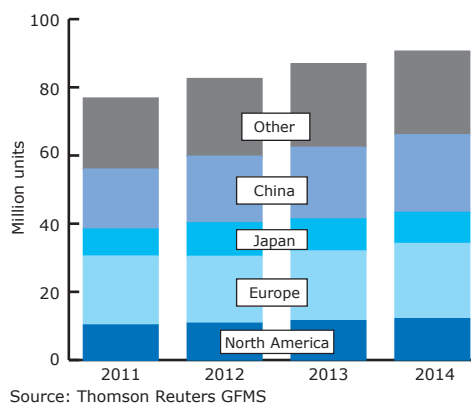
6.2 Autocatalyst

Overview

Although much has changed since our previous report in June of this year, the thrust of our outlook for platinum and palladium demand in autocatalyst applications remains broadly similar. In short, the prospects for demand growth in platinum remain subdued, while the outlook for palladium continues to look a good deal more promising. As far as platinum is concerned, this lacklustre outlook owes much to the concentration of demand in Europe (almost 50% due to diesel's large share of vehicle production), where the prospects for economic growth are seriously threatened by the euro-zone sovereign debt crisis and the continuing lack of a credible policy response from the region's leadership. Meanwhile, although not without its challenges, palladium has a more diverse geographic footprint and is dominant in the very much larger gasoline sector where the bulk of production growth is expected to occur. That our outlook for autocatalyst platinum and palladium demand remains broadly similar despite the recent turmoil (although our estimates are somewhat lower), is due largely to the conservative assumptions in our previous forecast – most notably surrounding vehicle production in Europe but also rather circumspect over volume growth in North America and China.

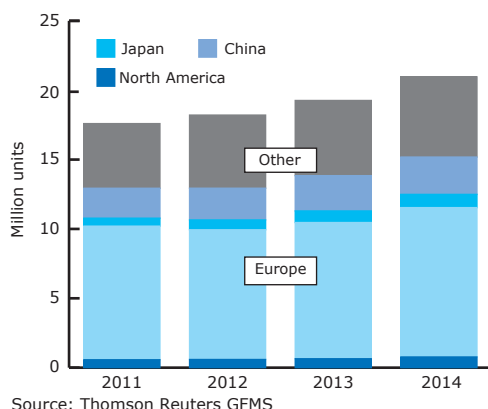
However, as the debt crisis continues to rage and so many questions remain unanswered, it would be foolish to present our current forecast as a definitive view. Instead, it should be regarded as a best estimate in a period of almost unprecedented uncertainty, wherein

Global Light Vehicle Production - All Fuel Variants

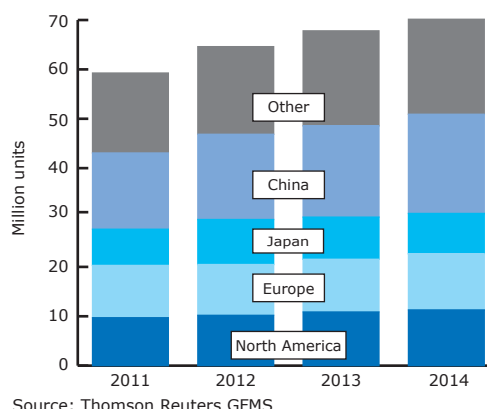




Global Light Diesel Production



Global Light Gasoline Production



the risks are most likely biased to the downside. That said, while the recent price falls in platinum and palladium (as well as in the markets in general) may resonate with the situation in late 2008, it is difficult to visualise this as a prelude to anything like the scale of auto industry melt-down which occurred in the developed economies at that time. That recession was characterised not only by its severity but the great speed at which it occurred, leaving massive unsold inventories and triggering widespread assembly-line closures in a desperate bid to conserve cash. Furthermore, that period also heralded the soon to follow bankruptcies of General Motors and Chrysler and a near-death experience for Ford as well. Difficult as things may eventually become, it is hard to imagine an outcome anything like as bad this time around, with the US auto sector having since been drastically restructured, inventories brought under control and cash balances in the industry as whole considerably improved and, in some cases, very substantially indeed.

Vehicle Production

While the demand profile for platinum and palladium in autocatalysts is little changed, there have been material revisions to our vehicle production forecasts which, in global terms, result in increased volumes relative to our previous assessment. However, with the exception of an upgrade to Japan's production in 2011, these increases are attributable to our "other regions" segment (mainly gasoline), where PGM loadings are generally less than the global average. Countering these derived benefits are reduced growth expectations for North America, where the economic recovery is increasingly fragile, while production growth in China has also been scaled back in the continuing absence of state-funded incentive schemes; these having been the principal driver of surging production

in both 2009 and 2010. Meanwhile, the situation in Europe is obviously problematic, not only due to policy uncertainties over sovereign debt, but also the effects of widespread austerity measures over public spending and taxation. While this has yet to impinge on growth expectations for 2011, the European economy is now so severely threatened that a decline in the region's vehicle production in 2012 seems to us to be inevitable (estimated at nearly 3% year-on-year). This contrasts with our previous assumption of a continuing if hesitant recovery next year and forms a reduced base (and hence lower volumes than previously visualised) for the anticipated recovery in 2013 and 2014.

Platinum Demand

In terms of global demand for platinum in autocatalysts, having registered a 16% improvement in 2010, we anticipate a very slender increase this year (only 1.6%) to a world-wide total fractionally in excess of 3M ounces. This muted increase in 2011 is due to a combination of circumstances, including the earthquake and tsunami in Japan earlier this year (although the auto industry's speed of recovery has been impressive), relatively subdued vehicle production in Europe (within which diesel's share is up only marginally) and the debilitating effects of continuing substitution in favour of palladium.

Moving forward, despite the situation in Europe, we expect a marginally better result for autocatalyst platinum demand in 2012 (up 2.4% globally), aided by a sharp recovery in Japan (although subsiding thereafter) and improved demand from the diesel sector in China, enabled by the anticipated availability of low sulphur diesel fuel (this has been long awaited and frequently delayed). Thereafter, we expect better outcomes in 2013 and 2014, when demand



Platinum Autocatalyst Fabrication					Palladium Autocatalyst Fabrication				
(000 ounces)					(000 ounces)				
	2011	2012	2013	2014		2011	2012	2013	2014
North America	435	442	463	507	North America	1,266	1,335	1,405	1,503
Europe	1,496	1,388	1,444	1,604	Europe	1,566	1,557	1,632	1,827
Japan	306	346	324	316	Japan	733	924	888	873
China	184	249	378	399	China	1,108	1,312	1,416	1,518
Other regions	613	681	712	766	Other regions	815	940	1,020	1,083
Total	3,035	3,107	3,321	3,591	Total	5,488	6,068	6,361	6,803
y-o-y %	2%	2%	7%	8%	y-o-y %	4%	11%	5%	7%
Source: Thomson Reuters GFMS					Source: Thomson Reuters GFMS				

is expected to improve by an average 7.5% in each year. This profile should enable platinum demand of 3.6M ounces in 2104; the final year of this forecast. While this envisaged outcome in 2014 is a considerable improvement over current levels of demand (and very much better than the 2.6M ounce low-point of 2009) it remains well adrift of record high of 4M ounces in 2007; a level which could well take several more years to regain.

Within this profile, it is worth noting a distinct **lack momentum for platinum in light duty applications**. This is especially troubling as light duty (principally diesel but including some demand in gasoline as well) currently represents 90% of platinum's global demand base in autocatalysts. This apparent inertia can be attributed to three major factors. Firstly (as we have highlighted already), lacklustre production growth in Europe which, as a result of diesel's disproportionately large share (some 48% of production), accounts for nearly half of global autocatalyst platinum demand. Secondly, diesel's share of light vehicle production is largely static, not only in Europe but elsewhere as well. This is particularly relevant when contrasting platinum's future prospects with its more illustrious performance over much of the previous decade. Since the late 1990s and through most of the last decade, diesel's market share growth in Europe (and the near exclusive use of platinum in this application) was the major driver of global platinum demand in autocatalysts. However, diesel's share growth has dissipated since 2007 when its' penetration of Europe's production first achieved the level we see today.

Thirdly, continuing substitution of platinum by lower cost palladium, both in diesel (partial but nonetheless damaging) and gasoline applications. To illustrate this; when global demand for platinum in autocatalysts peaked in 2007, its dominance of the key light duty

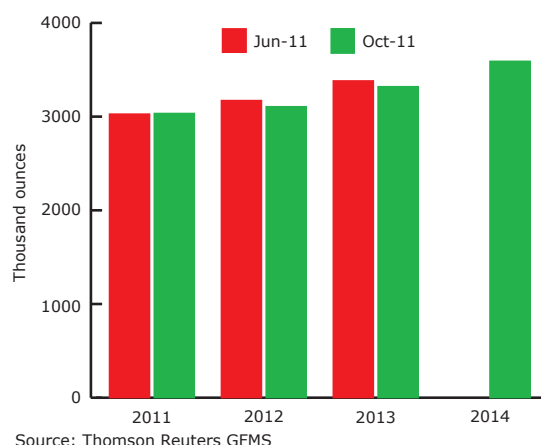
diesel sector in Europe was around 90%, with the use of palladium (the remaining 10%) in its infancy. As of this year, platinum's share of this application will have fallen to less than 75%, with palladium's penetration having increased correspondingly over a quarter. While the pace of substitution is expected to slow over the forecast period, it will continue to debilitate platinum demand in light duty, given the apparent lack of impetus from either volume or market share growth.

Consequently, the growth we are forecasting for platinum demand in autocatalysts is almost entirely attributable to the **heavy duty sector**, mainly on-highway applications but also in the emerging off-road sector (dubbed non-road mobile machinery, or NRMM). In the developed economies, this is driven by still recovering sales of on-highway commercial trucks (since sales in 2009 fell to less than half normal levels), supported by tighter emission standards, including EPA 2010 in the United States and Euro VI in the European Union (EU) from 2013. Meanwhile, there are considerable growth opportunities in a number of other markets where autocatalyst enabling emission standards are being adopted for road-going trucks. These include China where the availability of low-sulphur diesel fuel is anticipated in 2012 (this has long been delayed), enabling the nationwide roll-out of Euro IV compliant vehicles 2012.

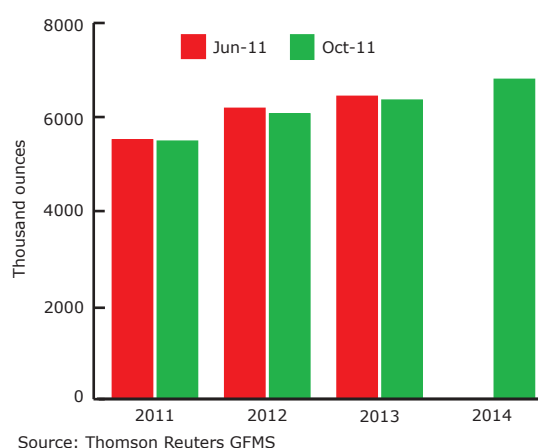
Meanwhile, autocatalyst enabling regulations have been adopted for NRMMs in the United States, EU and Japan (phasing in between 2011 and 2014) and will make a material contribution to platinum demand. However, platinum demand in heavy duty sector (on and off-road) is not immune to the threat of substitution and the use of palladium in these applications is also expected to rise (although platinum is expected to remain dominant).



Platinum Autocatalyst Demand



Palladium Autocatalyst Demand



Palladium Demand

Meanwhile, the prospects for palladium demand in autocatalyst are more encouraging. Given palladium's dominance in gasoline applications (three-quarters of global vehicle production), its demand base is significantly larger (1.8 times platinum's in 2011), more broadly based, and is expected to grow quite strongly over the forecast period. Having staged a more than complete recovery in 2010 (up 30% after a recession induced fall of 11% in 2009), autocatalyst demand for palladium is expected to improve by 4.4% to a total of nearly 5.5M ounces in 2011. Although moderate at first glance, substitution gains (mainly in diesel) are enabling palladium demand growth to exceed this year's anticipated rise in global gasoline vehicle production; the level of which has been restricted by Japan's natural disasters and China's post-incentive retreat to single-digit production growth for only the second time in ten years. Looking ahead to 2012, demand for palladium should improve considerably (up 10%) as Japan's auto industry completes its recovery and as double-digit production growth (albeit low field), resumes both in China and the "other regions" (the vast majority being palladium friendly gasoline variants). Thereafter, we expect demand growth for palladium to stabilise from 2013 at an average 6% per annum over the final two years of this forecast, lifting palladium demand in autocatalyst applications to a total of 6.8M ounces in 2014.

As we have indicated in this and in many previous reports, autocatalyst demand for palladium has enjoyed

considerable benefits from substitution gains at the expense of platinum. However, while opportunities remain for palladium to increase its penetration of individual applications, the gains that palladium can expect to enjoy over the horizon of this forecast will be considerably lower than those of recent years. In the light duty diesel sector, the introduction of bi-metallic platinum/palladium catalysts (where platinum remains dominant) first began in 2006, with diesel autocatalysts prior to that point having been almost exclusively platinum based. However, driven by the availability of ultra-low sulphur diesel fuel, such catalysts have since been rapidly adopted in the developed economies (those with leading edge emission standards) and in particular in Europe, the world's largest light diesel market. This has lifted palladium's penetration of the global light diesel sector from virtually zero to an estimated 20% as of last year. While there is every prospect for palladium's presence in this application to increase (at platinum's expense), this is likely to be more gradual over the next few years, pending the availability of ultra-low sulphur diesel in places like China and most likely, new catalyst formulations geared to Euro 6 compliance in Europe from late 2014. Meanwhile, palladium has also rapidly increased its domination of gasoline autocatalysts, cutting platinum's share of this application in half over the last five years and reducing its presence to marginal levels. While there is still scope for palladium to displace the residual usage of platinum, the majority of this replacement has already occurred and we are now entering a period of diminishing returns.



6.3 Jewellery

Platinum

In 2010, global platinum jewellery demand fell by 17% to 1.89M ounces. The decline was driven by losses in the price sensitive Chinese market, which accounts for almost two-thirds of global fabrication. This year, however, we expect to see a recovery of some 15% in China, due primarily to consumer acceptance of higher platinum prices. This will drive the 9% rise in global demand, after which three further years of gains are forecast. It is the growth in Chinese fabrication that accounts for the bulk of these gains, as continued economic growth and platinum's weakening price profile over the next three years should support growth there. This, combined with gains in North America and Europe (and a largely neutral result in Japan), will see global demand reach a total of around 2.7M ounces by 2014.

China

Looking back at Chinese platinum jewellery fabrication over the last decade, we can see extraordinary swings in jewellery output. After reaching a peak of 1.67M ounces in 2002, offtake halved to no more than 835,000 ounces four years later, before rebounding strongly in late 2008 to then achieve a seven-year high of just above 1.6M ounces in 2009. Thereafter, however, demand fell back last year, dropping by 26% to less than 1.2M ounces.

The key factor behind this dramatic change in jewellery fabrication has been the trend in platinum prices, as a massive increase in prices in the middle part of the 2000s saw the competition from more affordable white K-Gold and palladium jewellery impact platinum's market share. Not surprisingly, after the collapse of the platinum price in late 2008, demand immediately recovered, on the back of greater price attractiveness to consumers as well as an improvement in the jewellery trade's profit margins.

The price sensitivity of platinum jewellery demand is also clear from intra-year developments in 2011. During the first eight months of this year, against a backdrop of elevated platinum prices, demand remained broadly flat year-on-year. Moreover, a reluctance to be exposed to the high price of platinum has led some fabricators to shift in favour of 24-carat gold jewellery, whose sales have experienced a rapid increase in recent years. In contrast, following the steep correction in platinum prices in the second half of September, output has since recovered strongly. For example, turnover on the Shanghai Gold Exchange

(the only official source of VAT-free platinum in China) posted its highest monthly figure in September. Meanwhile, our latest information from various trade sources suggests that unofficial flows of platinum from Hong Kong to mainland China have also picked up over the last few weeks.

Looking ahead, as we expect the price to trade below \$1,700 over much of the fourth quarter of this year, further gains in jewellery demand (at least at the wholesale level) are possible before year-end, which largely explains an upward revision in our forecast for 2011 compared with the June forecast. Nevertheless, it is important to point out that, while we are fairly positive regarding jewellery demand in the near term, we would caution that the notable strength of jewellery demand seen during late 2008 and 2009 is unlikely to be repeated in the coming months.

To a large extent, this is because the recent rebound in jewellery demand has been mainly driven by retailers' re-stocking. In contrast, sales at the retail level have remained broadly flat (except during the long holiday in October), as shops have been reluctant to cut their price points. For example, although the platinum price dropped from a high of over RMB 380/g in early September to around RMB 310/g by early October, many retailers decided to leave prices unchanged, or in some cases lower them by a mere RMB 10-20/g, with Pt 950 jewellery therefore still being currently sold at RMB 450-500/g across the country. In addition, unlike a few years ago, when platinum was generally perceived as the "metal of choice" among consumers, there has been growing enthusiasm for 24-carat gold jewellery (which once was viewed as old fashioned) in the general public, particularly in the context of surging inflation in the local market, as plain gold jewellery was often bought as a safe haven asset.

Once the restocking process has largely been completed, we would argue that platinum jewellery demand could see a noticeable slowdown, particularly given that the metal price by that time will be back above \$1,700. As such, demand is forecast to post a relatively modest 8% increase in 2012 before weaker prices from 2013 onwards provide another strong boost to jewellery demand, which will take it to a new all-time-high of close to 2M ounces in 2014. Aside from lower prices (at least in 2013 and 2014), jewellery fabrication should also be helped by further gains in disposable incomes over the forecast period, itself a result of a broadly stable local economy. Related to this is that demand will benefit from a rapid expansion



of the retail network across China and an increase in platinum's penetration in smaller cities (targeting the fashion conscious youth demographic). To illustrate the scale of this growth, our field research has highlighted one example of just one retailer who plans to open around 200 shops per annum over the next few years. Furthermore, from discussions with Chinese retailers we also conclude that there is a great deal of confidence across the country's jewellery market, which is supported by rising affluence and robust economic growth in China.

Tuning to net jewellery demand, while the rising price earlier this year encouraged some modest growth in jewellery scrap, volumes remained below their peak seen in the first half of 2008 as the absolute price was not sufficiently high to generate a new wave of recycling. With prices falling by over \$350 in late September, scrap also fell sharply. Thereafter, as prices are expected to trade below their historic peak over the next three years, scrap should remain generally weak, which suggests that demand excluding scrap will post slightly higher growth compared with the forecast trend in gross jewellery demand.

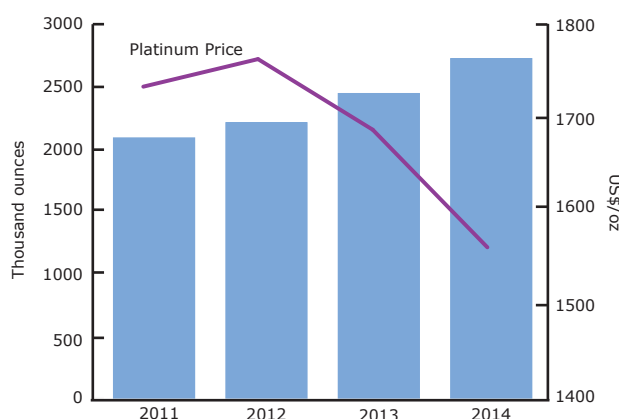
Japan

Last year, gross platinum jewellery demand in Japan is estimated to have fallen by 3% to just over 260,000 ounces. This year, demand is forecast to fall once more to around 250,000 ounces, before settling at this lacklustre level in both 2012 and 2013, which is just over half the level seen in 2007. Since the June report, the forecast series has remained unchanged.

The first obstacle to platinum demand in the Japanese market is the fact that consumer confidence remains weak. The jewellery market as a whole has been in decline for a number of years; not only has the recent credit crunch exacerbated this trend, but so too has the more recent tragedy of the major earthquake and tsunami (and ensuing radiation crisis), which have damaged the country's export capabilities, undermining its economic growth prospects. Moreover, with rising concerns over the world's sovereign debt crisis and the weaker outlook for the global economy, demand for luxury goods, including jewellery, is very likely to be reduced.

Secondly, the demographics of platinum jewellery-buying consumers have changed. Mature shoppers are no longer spending at the rate that they used to, while the younger generation tends to shun traditional luxury and designer goods. Such attitudes are unlikely

Global Platinum Jewellery Fabrication



Source: Thomson Reuters GFMS

to change within this forecast period, and are likely to add further downward pressure to demand. Third, certain trends that evolved last year to reduce the platinum content in jewellery are likely to remain in place throughout the forecast period. In general, delicate, lightweight designs have gained in popularity over heavier pieces (which have kept metal content and therefore prices down). In addition, combination products (typically platinum and gold) have also gained display space in retail stores.

The bridal sector (which forms the core of Japanese platinum demand) is set to remain broadly flat over the forecast period. Platinum holds considerable cachet in Japan, and is considered to be the metal of choice for bridal jewellery, which has ensured that demand has remained relatively stable in the face of retail price rises and economic weakness. That said, there is more downside risk than upside potential for bridal jewellery. In particular, there is little scope for increased sales of wedding rings, as the market is already largely saturated with platinum, while the number of marriages is forecast to decline.

With regard to engagement rings, a recent trend has been for couples to forgo these entirely (due to economic considerations). With the Japanese economy likely to shrink this year and struggle thereafter for sustainable growth, this adverse trend may continue. In addition, substitution in favour of white gold, which is relatively cheaper, remains a persistent threat to platinum jewellery.

However, in the final year of the forecast, jewellery demand is projected to rise by 3% to 255,000



ounces. This is due to a general improvement in the global economic environment, as well as the Japanese recovery in the aftermath of the earthquake and tsunami, both of which should improve the country's economic growth and, as a result, consumer confidence.

This year, platinum jewellery scrap is forecast to fall by 15% to just over 300,000 ounces. This is largely being driven by Japanese yen appreciation (carrying on from last year), which implies that the rise in dollar platinum prices will be fairly subdued in yen terms. Thereafter, we expect scrap volumes to remain fairly stable, around 300,000 ounces, over the remainder of the forecast period. This is because yen prices are expected to stay relatively steady and at a level which is unlikely to stimulate a rise in selling. Scrap volumes will, however, remain at elevated levels in historical terms, due to a highly developed scrap collection infrastructure, as well as greater public awareness of collection outlets.

Europe & North America

Western platinum jewellery demand fell a substantial 40% from 2000 to 2009. This was largely price driven, with damage worst in the more price sensitive mid-tier fashion segment. Recent economic problems also contributed, with the 2009 fall being more than twice as steep as the 2001-08 average. Last year saw a partial recovery thanks to an improved economic backdrop and the end of destocking by the international jewellery and watch brands. The overall rise occurred despite higher platinum prices (which hit bridal sales) and retailers' cautious approach to stocks.

Despite still high prices for platinum on a historical basis, the emergence of a notable discount to gold means offtake should grow modestly this year and next. The picture within this will be mixed, as suggested by the 21% year-on-year fall for UK hallmarking for January-September this year. A retreat in prices and economic normalisation should then assist consumption from 2013 onwards but a return to a substantial platinum:gold premium in 2014 means growth could grind to a halt. Prices also stay sufficiently high to rule out the return of the mid-tier, explaining why we foresee no return to the fabrication levels of 10 or so years ago. Ongoing substantial gains for the high end brands, stemming mainly from emerging market sales growth, will most benefit Europe and this should enable its growth to outpace North America's.

Our western jewellery estimates have been raised slightly since the last report thanks to lower forecast

prices for 2011 and 2012, the emergence and forecast persistence of a platinum:gold discount and to better than expected top end sales. Swiss watch hallmarking for instance has swung sharply from a year-on-year decline of 17% for January-June to a 4% increase for January-September. Total US jewellery sales have, in contrast, proved more disappointing than expected.

After the heavy fall in 2010, mainly in North America, platinum jewellery scrap should slide steadily to 2014, as selling back by consumers and the trade declines despite firm prices. This means net demand grows slightly quicker than gross, although the difference is slight. Changes since the June report mainly involve a slight increase to North American scrap.

Palladium

Last year, palladium jewellery fabrication fell by 27% to 809,000 ounces, a seven-year low. This was led by the 40% decline in Chinese fabrication, itself mainly driven by the negative sentiment at the retail and consumer level, as well as the rising palladium price. If Chinese demand is excluded from the total, the decline was a more modest 7%, with demand in North America and Japan decreasing by 7% and 1% respectively; a near 10% increase in European demand helped to partially offset these developments. Looking ahead, further losses in China are expected to account for the bulk of the losses in the global total, through to 2014. This, combined with steady declines in almost all other markets, will outweigh expected gains in Europe, where palladium's use in gold jewellery alloys is set to increase over the forecast period. By 2014, global demand is projected to fall to around 550,000 ounces, almost half the level seen in 2009.

China

Demand for palladium jewellery suffered a heavy loss of some 40% to 395,000 ounces last year, a decline which in fact had first emerged during the second half of 2008. Not only did the 2010 total represent the lowest level since the introduction of carat palladium jewellery in China (in the latter part of 2003), but it represented less than 40% of its peak level in 2005 (with China's share of global jewellery fabrication falling from 76% to 49% over the same period). Looking ahead, despite a small upward revision to the forecast series, our view remains little changed compared with the previous report. In other words, the downward trend will continue throughout the forecast period, although the extent of the decline will be less pronounced by 2014 (compared with the pace of the losses during 2012-13).



Platinum Jewellery Fabrication					Palladium Jewellery Fabrication				
(000 ounces)					(000 ounces)				
	2011	2012	2013	2014		2011	2012	2013	2014
North America	218	222	228	221	North America	116	109	103	97
Europe	209	221	227	234	Europe	147	156	166	178
Japan	248	248	248	255	Japan	43	43	43	45
China	1,360	1,464	1,684	1,954	China	322	260	215	190
Other regions	37	38	39	40	Other regions	70	60	50	40
Total	2,072	2,193	2,425	2,704	Total	698	629	577	549
y-o-y	9%	6%	11%	11%	y-o-y	-14%	-10%	-8%	-5%
Source: Thomson Reuters GFMS					Source: Thomson Reuters GFMS				

As we have discussed in previous reports, the hefty decline forecast over the next three years is largely due to long standing structural problems in the industry, including unfavourable tax policies and the low awareness of palladium jewellery among the general public. As for the former, unlike gold and platinum, palladium is not traded on any local exchange and is therefore subject to VAT on imports. This anomaly is therefore acting as a disincentive for some retailers to offer palladium jewellery. For example, many of the largest retailers in the first tier cities have decided it is not worth promoting palladium jewellery while such challenges persist.

In addition, the absence of officially quoted palladium prices has also resulted in a lack of price transparency for consumers and the lack of a true two-way market for the metal. Even though some retailers of palladium jewellery have begun to offer buybacks (and often at reasonable spreads), these still represent a distinct minority. Overall, these factors present major obstacles in establishing palladium as a valid choice in the eyes of the jewellery buying public, compared to 24-carat gold and platinum, where the consumer can always cash-in their items.

Related to this is a lack of public awareness of palladium jewellery, this being the result of limited promotional campaigns, which have further undermined the jewellery trade's interest. Clearly, palladium still needs to overcome a high level of consumer ignorance about its intrinsic value and the fact that it is a durable and viable alternative precious metal for fine jewellery. As a result, since late 2008 the situation has deteriorated still further, with demand for platinum having rebounded (in 2009 at least), and 24-carat gold jewellery sales maintaining a strong growth path since

that time. As a result, many retailers decided to reduce their palladium stocks or even, in some cases, to stop selling palladium jewellery altogether.

Returning to the details of our forecast, our recent trip to China in September has confirmed that demand has continued to fall this year. In addition, despite a sharp fall in prices more recently, demand for palladium jewellery has remained generally weak over the past few weeks, mainly driven by the persistence of the aforementioned issues. Given there is unlikely to be any major marketing promotion in the near term, we do not expect the current pessimism regarding palladium jewellery throughout the value chain to change in the near future.

Overall, therefore, we expect a 19% fall in demand in both 2011 and 2012, followed by a smaller decline of 17% and 12% in 2013 and 2014 respectively. It is of note that although we have revised upward our forecast for 2011-2013 slightly compared with the previous report, it must be seen in the context of the extent to which jewellery demand had already declined over the last two years. Indeed, by 2014, palladium jewellery offtake is still expected to drop to no more than 190,000 ounces, to only 18% of the peak level seen in 2005. As a result, palladium jewellery will have shrunk to a niche market by that time, with sales concentrated in certain rural areas where incomes are relatively low and where consumers see palladium as an affordable alternative to platinum.

Finally, it is worth mentioning that we remain of the view that the volume of jewellery scrap will trend higher over the forecast period. Key to this is the current pessimism relating to palladium jewellery, which could encourage more retailers to sell their stocks back to



manufacturers over the next three years. As such, the fall in net demand will be even greater than suggested by the trend in gross palladium jewellery fabrication.

Japan

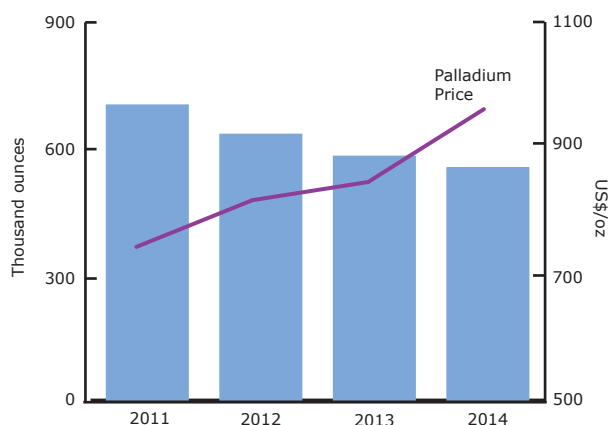
In Japan, palladium is used mainly as an alloying ingredient for both white gold and platinum jewellery. In 2010, demand fell by just 1% to around 48,000 ounces. Over the forecast period, offtake is expected to remain subdued, a reflection of the lack of growth in the jewellery market as a whole. As for carat palladium jewellery, there has in general been little effort to promote this as a jewellery metal in its own right, although there have been some tentative attempts to launch the product. In general, consumer awareness of palladium as a precious metal is still lacking, which has translated into a low level of sales. Furthermore, some retailers and manufacturers have raised concerns that the somewhat dark appearance of the metal (compared to platinum or rhodium-plated white gold) may limit its success, as will platinum's entrenched position as the jewellery metal of choice.

Europe & North America

Palladium jewellery demand in western markets falls into two distinct areas. The first is the metal's use as an alloy in white gold (mainly top end pieces in Europe). Mass market items (chiefly the price sensitive parts of the US market and export-oriented Italian output) in contrast mainly use nickel as the 'whitener' (with the exception of earrings). A steady move away from nickel on allergy grounds and the growing share of high end pieces within the total means the proportion of white gold using palladium has been rising. However, the fall in western fabrication in recent years means the weight of palladium in alloy has not risen. The second area is carat palladium, normally of a 950 fineness. This has grown markedly since the mid-2000s, although the levels now reached remain fairly modest.

Despite these early gains, carat palladium's prospects are mixed. Europe should still enjoy decent growth, chiefly in the bridal sector due to high and sustained price differentials versus platinum. Germany and France look to offer good potential, while hallmarking in the United Kingdom has assisted demand there. This may seem contradicted by the year-on-year fall in recent quarters' hallmarking figures but this is thought due to earlier figures being boosted by trade stocking, with underlying UK sales still growing. In contrast, nascent interest in Italy seems to have faded and we remain less than confident as regards prospects for carat palladium sales in the US market; most of

Global Palladium Jewellery Fabrication



Source: Thomson Reuters GFMS

its major retailers have failed to introduce this and the independents' rush of enthusiasm has distinctly cooled, often citing consumer ignorance. In addition, it is doubtful that the top end international watch and jewellery brands will adopt palladium beyond the occasional foray into its use.

We expect that the use of palladium in white gold might initially weaken due to price damage and we cannot ignore the ongoing shift in fabrication to low labour cost producers. Nonetheless, the decline will be partly countered by the ongoing migration from nickel to palladium, and this process would be greatly assisted from 2013 should forthcoming tighter EU legislation on nickel use generate material substitution to palladium.

Our estimates for western palladium jewellery demand have been trimmed a little since June, despite the above substitution possibility and the cut in forecast palladium prices. This is mainly down to the deterioration in the outlook for all jewellery sales, particularly in North America.

Palladium jewellery scrap is forecast to rise a fraction in 2012 but then ease modestly in 2013 and 2014. The initial gains stem in part from inventory melt by the trade and manufacturers in the face of rising metal prices and disappointing sales, particularly in the United States. This adjustment phase should then end and we should also see a falling contribution from the small amount of white gold that is recycled. The amount of carat palladium coming back from consumers should, however, remain slight for the whole forecast. Our scrap figures are essentially unchanged since June.



6.4 Industrial Demand

Summary

This section covers platinum and palladium demand in industrial applications (net of recycling), including the chemicals industry, electronics, glass production, oil refining and the dental sector. After the downturn of 2009, which hit platinum particularly hard, demand in these areas staged a notable recovery in 2010. This was especially the case for platinum where demand rebounded by 50% to an estimated 1.85M ounces, having suffered a 31% drop the year before. In 2011, this level of platinum demand has been broadly maintained (up 1% to 1.88M ounces), indicating a continuation of the elevated glass sector demand we saw in 2010, that having been key to platinum's recovery.

Meanwhile, industrial demand for palladium is less dependent on major capital projects, incurring only moderate losses in 2009 of some 8%. With improved demand from the electronics sector in particular (this being the largest of palladium's industrial applications) these losses were quickly redressed in 2010 when overall demand increased by 10% to a total of 2.51M ounces. However, with electronics broadly stable this year, industrial demand for palladium is likely to rise only moderately this year, with overall demand in 2011 estimated at 2.56M ounces.

With industrial demand for both metals having returned last year to broadly pre-recession levels, we anticipate only minimal average growth over the whole of the forecast period. In the case of platinum, however, levels of net demand are expected to be rather erratic due to its heavy reliance on major capital spending projects and their timing, most notably in the glass

sector. To illustrate this point, in 2014 (the final year of this forecast), we anticipate overall demand for platinum industrial applications of 1.85M ounces; fractionally down on this year's level. This outcome is due, in the main, to considerably fewer expansions in glass that year, relative to the preceding two years (2012 and 2013) when overall industrial demand for platinum is expected to be around the 2M ounce mark. However, in the case of palladium, there are no such mitigating circumstances and the outlook over the whole of the forecast period is essentially flat. This is mainly attributable to the electronics sector and is the result of continued thrifting and substitution, together with increased recycling from old electronic equipment. The effect of this, together with a downward drift in the dental sector, results in overall demand for palladium in industrial applications of 2.59M ounces in 2014; little different from the 2.56M ounces we expect this year.

Sector Analysis

Demand for platinum from the **petroleum** sector is expected to remain broadly stable in 2011 at 200,000 ounces, with steady requirements for loss replacement and little by way of new capacity additions. Next year, however, the addition of new facilities, together with a greater number of catalyst change-outs (including postponements from 2011), are expected to boost platinum demand to a little over 250,000 ounces. However, this is expected to be short-lived, with demand for platinum trending lower thereafter to a below par 156,000 ounces in 2014. Weak as this end result may appear, this should not be viewed as the beginnings of a secular decline. Demand in this sector can be variable and if one looks at the average level of demand we are forecasting, this remains very much in line with historical averages, despite the apparent weakness in 2014.

Platinum Industrial Fabrication					Palladium Industrial Fabrication				
(000 ounces)	2011	2012	2013	2014	(000 ounces)	2011	2012	2013	2014
Chemical	463	464	556	491	Chemical	361	373	372	371
Electronics	238	262	250	247	Electronics	1,456	1,495	1,482	1,480
Glass	487	433	491	386	Dental	647	650	653	628
Petroleum	200	253	186	156	Other industrial	93	93	103	107
Other industrial	488	521	537	567	Total industrial	2,557	2,611	2,609	2,586
Total industrial	1,876	1,933	2,020	1,847	y-o-y %	2%	2%	0%	-1%
y-o-y %	1%	3%	5%	-9%	Source: Thomson Reuters GFMS				
Source: Thomson Reuters GFMS									

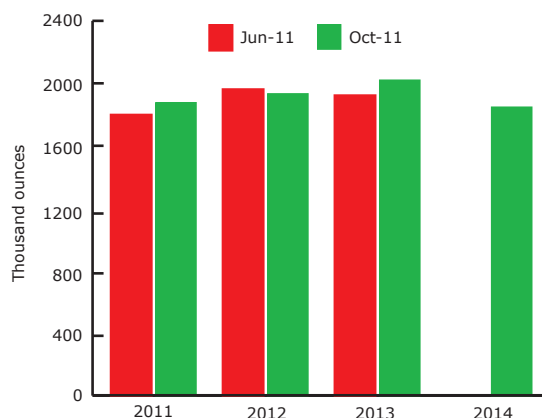


After a sharp decline in 2009, demand for platinum in the **chemical** sector enjoyed a marked increase in 2010 aided by the completion of a raft of new capital projects in the Middle East and China and a significant rise in capacity, most notably in paraxylene production. This was supported by stable demand for nitric acid production and increased consumption in speciality silicones. Overall, this enabled platinum demand to reach 463,000 ounces in 2010 – a record high in our data series. Despite some easing of capacity expansions, chemical sector demand for platinum is expected to remain resilient in 2011 and 2012 at close to this level, aided in part by nitric acid demand and most notably platinum-cured speciality silicones. In 2013, this should coincide with a further wave of capacity growth in paraxylene production, lifting demand above 0.5M ounces for the first time, before a moderate easing in 2014.

Meanwhile, after some improvement in 2010, demand for palladium in the chemicals sector is expected to continue its recovery in 2011 and 2012, stabilising at around 370,000 ounces in the period 2012 to 2014. While this remains well short of peak demand during 2006 and 2007, it is broadly in line with the five year average ahead of the recession in 2009. Demand for palladium in chemicals applications is influenced strongly by the schedule of capacity expansions in bulk intermediates, wherein products such as purified terephthalic acid (PTA) and vinyl acetate monomer (VAM) feature prominently.

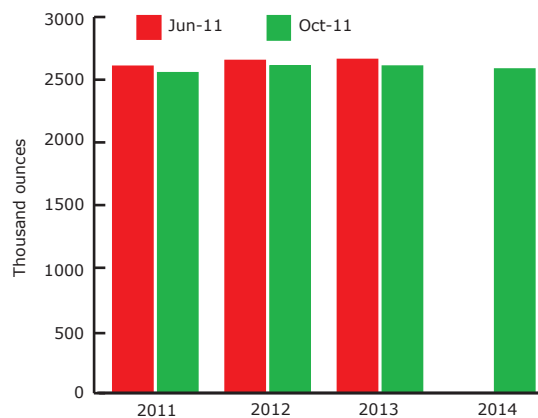
The largest component of palladium demand in industrial applications is **electronics**. After a short-lived (if severe) downturn in late 2008/early 2009, palladium demand in electronics regained its growth trajectory in 2010, exceeding 1.44M ounces on a net basis. This was aided by a variety of factors including improved sales of consumer electronic goods (for example handsets and personal computers), higher vehicle production (where electronics content is rising) and the increased use of palladium in electroplating as a less expensive alternative to gold. However, with sharply increased palladium prices, we anticipate negative implications for net palladium demand in electronics going forward. This includes renewed emphasis on thrifting (as well a more intensive focus on the use of less expensive, alternate materials) and is likely to limit demand growth in important areas such as multi-layered ceramic capacitors (MLCCs). In these devices, we expect that silver:palladium ratios will be adversely pressured, while cheaper nickel based alternatives (which have improved in terms of technical performance) may well achieve wider acceptance. In consumer electronics in particular, the impact of higher palladium prices (relative to the historical average) is magnified by declining average selling prices for the final product, thus magnifying the need for cost economies. Furthermore, elevated palladium prices (in conjunction with gold prices) improve the economic viability of recycling end-of-life electronic goods, further suppressing the outcome for net palladium demand in this sector. As a result, we expect net demand for palladium in electronics to remain in the 1.4M to 1.5M

Platinum Industrial Demand Forecast Revisions



Source: Thomson Reuters GFMS

Palladium Industrial Demand Forecast Revisions



Source: Thomson Reuters GFMS



ounce range over the forecast period and anticipate a 2014 total of 1.48M ounces.

As far as platinum demand in electronics is concerned, the dominant factor is its use in hard disk drives (HDDs) where it forms part of the data storage media. Although HDD shipments rose by an impressive 16% last year, demand growth for platinum in this application was a less impressive 6%, reflecting downward pressure on average platinum loadings alongside the increasingly prominent use of ruthenium; factors which will continue to constrain platinum demand over the forecast period. Looking forward, while HDD shipment volumes are expected to continue trending higher (apart from an interruption this year), underlying growth is likely to be considerably less than the double-digit rates often seen over recent years. In part, this reflects maturing demand but is also symptomatic of rapid changes in mobile computing (the main driver of recent HDD volume growth) brought about by the explosive growth of tablet computing and smart phones. Where mobile internet access is the primary requirement, such products are more convenient and their increasing popularity is a drain on demand for HDD based netbook and laptop computers.

In 2011, demand for platinum in HDDs will be adversely impacted by the extensive flooding in Thailand (the world's leading HDD manufacturing centre), the effects of which are unfolding as we write. Already, substantial factory closures have been announced and there are major disruptions to the nation's transport network, power supplies and other infrastructure. Although it will be some time before the consequences of this are fully known, we have cut our fourth quarter 2011 estimate for global HDD shipments quite severely, resulting in a moderate decline for this year as a whole. However, we are assuming these losses will be fully recovered in 2012 (along with previously assumed growth), although the effect of this will then unwind the following year, postponing the re-emergence of a normalised growth trend until 2014. Inevitably, all of this produces a rather erratic profile for platinum demand in electronics over the forecast period.

Finally, the use of palladium in dental applications (net of recycling) was unaffected by the recent recession and remained broadly stable in 2010 despite significantly higher prices. Demand in this area is supported in the main by state medical insurance in Japan (the world's largest user of palladium containing dental alloys) and palladium's cost advantage over gold – its' competitor as well as co-product in the

dental arena. However, although these aspects remain favourable, we see little scope for palladium demand growth in dentistry over the forecast period. The cost of such treatments has risen enormously over recent years, due especially to rising gold prices but also higher palladium prices; both metals featuring together in the alloy. Meanwhile, ceramic and plastic alternatives are cosmetically preferred and are likely to capture the majority of any growth arising from increased dental treatments. This leaves dental demand for palladium relatively flat for the time being, although the moderate reduction we are forecasting in 2014 could well prove to be the start of a secular downward trend.



7. Investment

7.1 Introduction

Since the mid-2000s, investment demand for both platinum and palladium has risen strongly, with the temporary exception of the massive sell-off that impacted the whole of the commodities complex. This growth in investment demand has been the most important driver of the increase in platinum and palladium prices in recent years. Behind this surge in investment have been several factors. First, we would mention the rise in investor interest in precious metals, particularly gold, with both platinum and palladium also benefiting from this trend. Second, support has come from the general growth in investment demand for commodities as an asset class, boosted by a weakening US dollar against most currencies in recent years.

Finally, both metals' supply/demand fundamentals at times have had a positive impact on investor sentiment. For example, supply concerns related to South African mine production triggered a sharp rise in speculative inflows into the platinum market in early 2008. In the case of palladium, the belief in convergence with platinum, as a result of the metal's ongoing substitution gains in autocatalyst fabrication, has also frequently generated interest, as have expectations of a near term cessation of Russian government stock sales.

As shown in the table below, investment demand for both metals is forecast to remain at healthy levels this year, the product largely of continued investor interest in the precious metals complex in the first eight months of the year, coupled with potential bargain hunting purchases following the recent price correction. Thereafter, however, we expect divergent trends in investor activity in platinum and palladium markets. As for platinum, investors will be required to absorb sizeable market surpluses from 2012 to 2014. This is a key reason why we believe the average platinum

price could fall well below \$1,600 in the final year of the forecast.

Palladium in contrast could generate gross deficits across the forecast timeframe, which implies that prices need to rise to a level at which some investors would be happy to sell their metal. The substantial fall in above-ground stocks these deficits will produce and the conclusion of Russian stock sales, will not only lead to a major increase in the palladium price but also result in the metal outperforming the other three main precious metals over the bulk of the forecast period.

7.2 Platinum

Investment has continued to play a central role in impacting platinum prices so far this year. While considerably lower year-on-year, investment demand for platinum on a net basis remained at healthy levels over the first eight months of 2011, helped by heightened investor interest in the precious metals complex, itself a result of ongoing loose monetary policies in major countries, a weak dollar and rising inflation fears. Moreover, certain investors bought platinum in the belief that it will play 'catch-up' this year following its underperformance relative to palladium during 2009 and 2010.

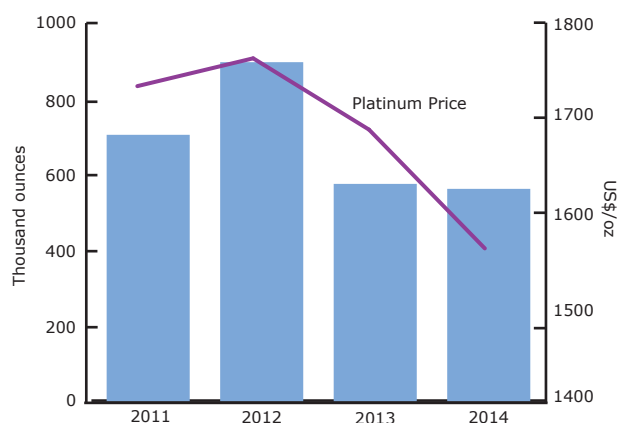
The brief rally in platinum during the autumn was also caused by a surge in speculative investor interest, especially after the gold:platinum ratio dropped below one in August, as traders took parity between the metals as a signal to buy platinum. This change in investor sentiment was clearly reflected by activity on Nymex, where net investor longs expanded by nearly 90% between end-June and early September.

Nevertheless, against a rapidly deteriorating economic backdrop, it soon became evident that platinum had

Platinum Investment 2011-2014					Palladium Investment 2011-2014				
(000 ounces)					(000 ounces)				
	2011	2012	2013	2014		2011	2012	2013	2014
Retail Investment	255	97	66	41	Retail Investment	99	81	69	62
Exchange Traded Funds	300	150	(200)	(100)	Exchange Traded Funds	(300)	200	100	(300)
Residual Surplus	410	726	768	655	Residual Surplus/(Deficit)	756	(275)	(395)	(138)
World Investment	965	973	634	596	World (Dis)Investment	555	6	(226)	(376)
Source: Thomson Reuters GFMS					Source: Thomson Reuters GFMS				

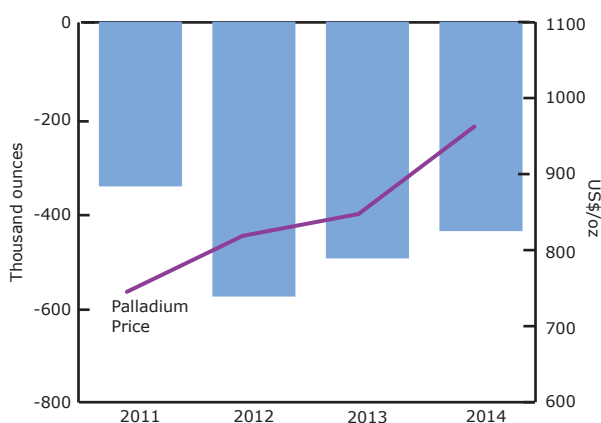


Platinum Gross Surplus & Price



Source: Thomson Reuters GFMS

Palladium Gross Deficit & Price



Source: Thomson Reuters GFMS

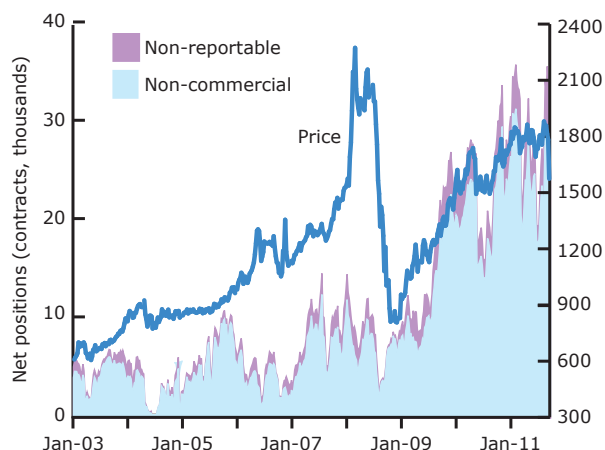
insufficient investor momentum to drive prices higher. This resulted in some profit taking in mid-September before the decline accelerated from 22nd September onwards, after the announcement of "Operation Twist" failed to restore investor confidence and subsequently led to aggressive selling of risky assets.

Looking ahead, we believe the scope for a quick rebound in investment demand will be limited. With a growing risk of a double dip recession in western countries and the persistence of the sovereign debt crisis, risk aversion will remain elevated in the short term. Related to this is growing worries about the health of the banking sector in Europe, which could cause liquidity to dry up in the coming months and force investors to shy away from risky assets. The potentially negative effect on platinum investment could be significant, particularly given the metal's heavy reliance on the European autocatalyst market.

However, it is important to point that, while there may still be room for some longs to be taken off and thus the potential for further price weakness, the downside risk has already been greatly reduced compared with a few weeks ago. In addition, a price elastic response from the jewellery sector in China and growing concerns over the long-term sustainability of mine production in South Africa should provide healthy support on any major price dip, if further long liquidation takes place in the coming weeks.

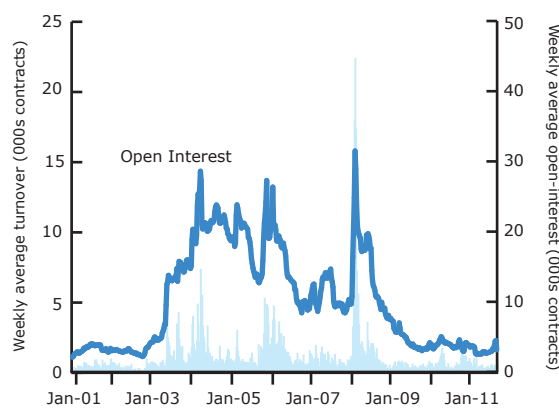
Once the more speculative long positions are closed out, we expect to see a gradual improvement in investor interest, which will keep investment demand as a whole in 2012 at similar levels to this year. Underpinning this is our expectation for continued zero to negative real interest rates, massive budget deficits and, in time, a resumption of the US dollar's decline. Indeed, with gold eventually breaking the \$2,000 barrier next year, this should provide a major boost to platinum, although gold's greater appeal as a

Tocom Platinum Futures



Source: Nymex, LPPM

Tocom Palladium Futures



Source: Tocom



safe haven and platinum's exposure to the lacklustre automobile sector suggest that the spread between the two could widen over the next few months.

Moving to 2013, we believe investment demand will weaken again, as signs start to tentatively emerge of a shift in favour of tighter monetary policies, which in turn should promote a secular rise in the US dollar. This, together with a gradual return to firmer global economic growth, is expected to encourage investors to shift their focus in favour of equity markets. In addition, as the metal is expected to remain in massive gross surplus over the entire forecast period, the unfavourable fundamentals should undermine investor interest, particularly when investor sentiment towards the precious metals complex changes.

Looking at the components of platinum World Investment, after a record high of net inflows into platinum ETFs last year, we expect there to be a slower pace of growth this year, although the net increase is still slated to reach 300,000 ounces. Thereafter, the increase in platinum ETF holdings is expected to be more modest but still positive, before falling prices result in investors beginning to liquidate their ETF holdings during 2013-14. It is worth noting that, compared with the previous report, we have trimmed our forecast for this year, reflecting lower investor appetite for the broad commodity complex in response to a bleak global economic outlook.

Turning to retail investment, the figure for 2011 has been revised upward considerably by around 200,000 ounces since the previous report. This is almost exclusively driven by a surge in investor interest in Japan over the past few weeks, a development that mirrored the final third of 2008 when a steep decline

in platinum prices triggered a wave of bargain hunting purchases. Thereafter, this component will follow a declining trend and then reach a seven-year low of just 41,000 ounces by 2014.

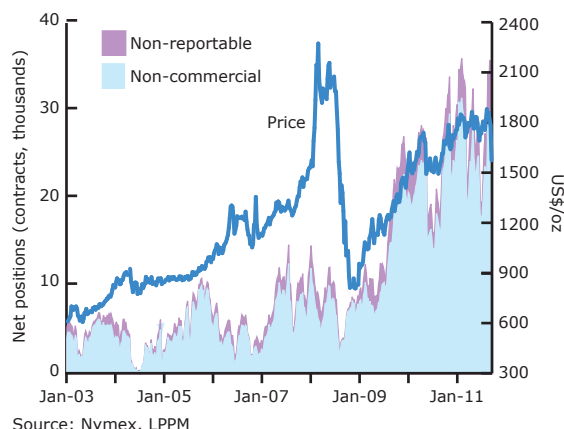
Finally, implied net investment (or the residual surplus) is forecast to rise strongly between 2011 and 2013, with the implied figure in the latter year over 150% higher than the level seen in 2010. Thereafter, the surplus will edge lower to 655,000 ounces in the final year of the forecast. These figures are, of course, balancing items, calculated from combining the other components of supply and demand. For this reason, the emphasis should be placed more on the year-by-year trends, rather than the absolute total. Therefore, it is worth highlighting the seven-year uninterrupted period, from 2008-2014, of residual surpluses in the platinum market, suggesting that investors in the over-the-counter (OTC) market need to be net buyers to absorb the surplus metal over the period.

7.3 Palladium

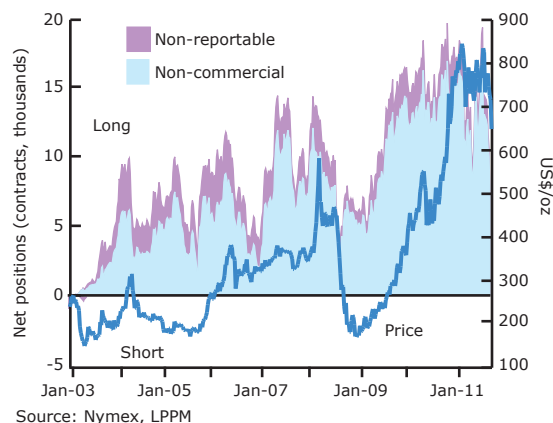
As discussed in the previous report, in contrast to 2010 when palladium was the best performer among the precious metals complex, the performance of the white metal this year-to-date has been fairly disappointing. To a large extent, this is because a number of positive factors that encouraged palladium investment had already been priced in. Investor interest was further dented by a major slowdown in Chinese car production, the Japanese earthquake and an escalation of the European debt crisis.

As with its sister metal, palladium is likely to struggle to gather much by way of sustainable forward momentum in the near term, particularly in the context of elevated risk aversion. However, while palladium may trade

Nymex: Platinum Total Net Positions



Nymex: Palladium Total Net Positions





sideways in the short term, we remains fairly bullish for the medium to longer term, due to an increasing tightness of the palladium market over the forecast period. Specifically, the healthier supply/demand backdrop owes much to a more rapid increase in autocatalyst demand, due to a faster pace of growth in regions that are dominated by palladium-rich autocatalyst systems and because of substitution gains at the expense of platinum in diesel. Moreover, we believe that the timeframe of this report will encompass the long awaited conclusion of Russian government inventory sales of palladium. After a continuation of elevated sales in 2010 and 2011, we think it likely that the volume of stocks that the authorities plan to dispose of will be close to exhaustion, with only residual amounts remaining for sale thereafter. While these may be fairly sizeable in 2012 (although much lower than historical norms), sales will most likely decline to minimal levels by 2013.

Indeed, as the gross deficit builds still further in 2012 (coinciding with a reduction in Russian government sales), this should enable palladium to bounce back above the \$800 by mid-2012. From 2013 onwards, although the broader climate for investment in precious metals may be showing signs of weakening, we anticipate an ongoing improvement in palladium's performance on the back of the aforementioned favourable supply/demand fundamentals. This should provide palladium with ample opportunity to trend higher, thus enabling us to forecast an annual average price of \$963 by 2014.

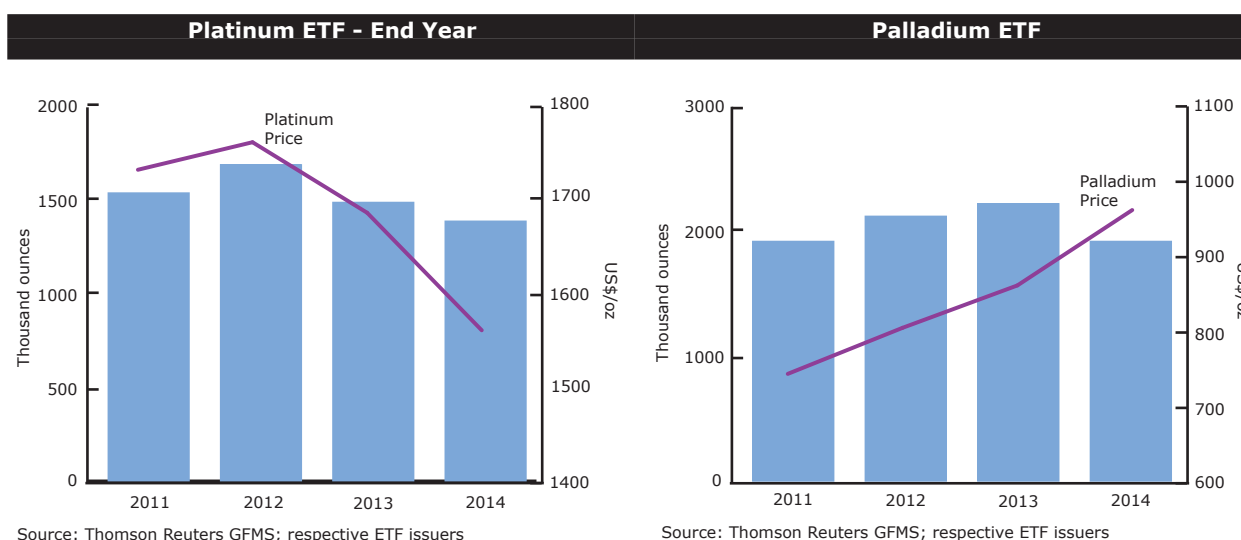
Furthermore, while platinum's recurring gross surplus implies that above-ground stocks will increase over the forecast period, palladium's growing gross deficits and a marked fall in Russian stock sales mean that above-

ground stocks will decline. Implicitly, this points to greater convergence of platinum and palladium prices by the end of the forecast period, with the spread between them nearly halving between 2010 and 2014.

Turning to the outlook for the individual components of investment demand, we now forecast a major decline of some 300,000 ounces in ETF holdings, mainly driven by profit taking or, more recently, the need to raise cash to meet margin calls. However, as prices have already fallen heavily, we expect to see some renewed buy-side interest from bargain hunters over the rest of the year. While further additions are likely (on a net basis) over the next two years, the pace will fall markedly, to just 100,000 ounces in 2013. As the price rises considerably towards the end of the forecast, we expect ETF redemptions to re-emerge in 2014.

Turning to global retail investment demand, the series has been raised slightly, reflecting somewhat higher than expected bar sales in North America, which appear to have picked up in recent weeks.

Finally, our residual surplus turned negative last year for the first time in ten years. In 2011, given the likely continuation of elevated Russian stock sales, coupled with a lack of interest in palladium ETFs, a residual surplus is expected to resume. Thereafter, as fabrication rises and Russian stock sales fall, we forecast a return to a residual deficit in 2012, with the figure continuing to grow in 2013 before edging lower during the following 12 months. In turn, this intensifies the need for the investment community to sell part of their holdings in order to provide the required palladium to satisfy fabrication demand, a development that is likely to see prices bid higher.





About Thomson Reuters GFMS

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